

EXHIBIT 37

NOT FOR PUBLICATION

**UNITED STATES DISTRICT COURT
DISTRICT OF NEW JERSEY**

MATSUSHITA ELECTRIC INDUSTRIAL
CO., LTD.,

Plaintiff,

v.

SAMSUNG ELECTRONICS CO., LTD.,
et al.,

Defendants.

SAMSUNG ELECTRONICS CO., LTD.,
et al.,

Counterclaimants,

v.

MATSUSHITA ELECTRIC INDUSTRIAL
CO., LTD.,

Counterclaim Defendant.

BROWN, C.J.:

Civ. No. 02-336 (GEB)

***MARKMAN* OPINION**

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INTRODUCTION

This matter comes before the Court upon the parties' request for claim construction in this patent infringement action. On January 25, 2002, Plaintiff Matsushita Electric Industrial ("MEI") brought suit against Samsung Electronics Co., Ltd. and its affiliates (collectively referred to as "Samsung") for infringement of three of their patents concerning semiconductor technology.¹ In particular, MEI asserts that Samsung's products infringe U.S. Patent No. 5,053,998 ("the '998 Patent"), U.S. Patent No. 5,375,095 ("the '095 Patent"), and U.S. Patent No. 5,475,648 ("the '648 Patent") (collectively referred to as "the MEI Patents"). Samsung counterclaimed alleging infringement of five of its patents relating to the same field of technology by MEI and some of its affiliates. Since then, four of the counterclaim patents have been withdrawn or dismissed, and only one counterclaim patent remains – U.S. Patent No. 5,751,048 ("the '048 Patent").

On March 7, 2005, the parties filed their Opening Claim Construction briefs with the Court. Thereafter, the parties submitted opposition briefs on April 11, 2005 and reply briefs on May 2, 2005. In their submissions, the parties included Joint Claim Construction Charts which identified the disputed claim terms, as well as stipulated constructions. (Certification of Michael V. Solomita, Ex. 4 ("MEI Patents Chart"); Certification of Gregory M. Hasley, Ex. 17 ("Samsung Patent Chart")). On January 19 and 20, 2006, the parties presented a tutorial on the relevant technology. The Court conducted a *Markman* hearing on February 9 and 10, 2006.²

¹ The case was initially assigned to the Newark vicinage. On December 6, 2005, the matter was reassigned to Chief Judge Garrett E. Brown, Jr. in Trenton.

² On January 27, 2006, Samsung filed a motion to seal certain confidential documents in connection with supplemental exhibits in support of Samsung's claim construction briefs. As discussed at the hearing on February 9, 2006, this Court did not grant Samsung leave to file supplemental exhibits during the technology tutorial on January 31, 2006. The instant motions

BACKGROUND

The subject matter of the four patents-in-suit relates to semiconductor memory devices. Dynamic random access memory, or DRAM, is an example of such a memory device. DRAMs have become one of the key products in the semiconductor industry. They allow for temporary storage of information that can quickly be accessed and recalled by a computer.³ DRAMs are capable of storing over 250 million bits of information on an individual semiconductor chip which typically can be as small as three-quarters of a square inch. With the increasing demand for faster computers and other electronic devices, efforts are consistently being made to increase the memory capacity and access speed of an individual memory device. Currently, DRAM chips that can store over 1 billion bits of information are in the early stages of development and production.

DRAMs typically consist of millions to billions of memory cells arranged in a two-dimensional array of rows and columns. Given this arrangement, each memory cell has a unique address based on the row and column of the matrix. The memory cell is composed of a capacitor, which stores electric charges, and a transistor, which allows for the release of these charges. On the vertical axis of an array, rows are formed by small wires known as word lines which are connected to the transistors of the memory cell. On the horizontal axis of an array, columns are formed by small wires known as data lines or bit lines which connect the capacitors of the memory cells. When

for claim construction have been fully briefed since May 2005, and the Court does not find a proper basis for receiving belated filings at this time. As stated at the *Markman* hearing, however, the Court will allow the supplemental filing of the prosecution histories for the '998, '095 and the '648 Patents, to which MEI does not object. Consequently, Samsung's motion to seal is denied as they do not pertain to the prosecution histories of these patents.

³ See Matsushita's Br. In Supp. of Its Claim Construction of the Matsushita Patents ("MEI's Br.") at 8-10; see also Samsung's Substituted Opp'n *Markman* Br. for the Matsushita Patents ("Samsung's Opp'n") at 2 n.3).

the central processing unit ("CPU") of a computer communicates with the DRAM chip to access and retrieve information, the CPU designates the unique address of the memory cell from which it wishes to retrieve the data.

Memory cells store information in the form of binary code, i.e., zeros and ones. In semiconductors, the binary value of zero or one is represented by an electrical signal or charge which is stored in the capacitor of the memory cell. For example, a binary value of one can be represented by a high charge and a binary value of zero can have either a low charge, or no charge at all. The process of retrieving binary values from memory cells is referred to as reading. Generally in a data read-out operation, a row address signal is sent to and decoded by the row decoder. '998 Patent, col. 1, ll. 19-36. The row decoder sends the decoded address to the word line driver which uses this information to activate the word line that corresponds to the row address. The electrical signals contained in the individual memory cells, which are connected to the activated word line, are transferred onto the bit lines.

These signals, however, are extremely small – so small that they cannot be understood by the rest of the DRAM components in that state. A sense amplifier is required to amplify, or strengthen, the signals so that they may be read as a zero or one. Additional circuitry involving differential amplifiers, sense amplifier drivers, and restore and drive signal lines act together to amplify the signals. These amplified signals are then sent to a data selector. The desired information in the data selector is transmitted to the data output buffer once the column address is determined by the column address buffer and column decoder.

DISCUSSION

I. Law of Claim Construction

The first step in a patent infringement analysis is to define the meaning and scope of the claims of the patent. *Markman v. Westview Instruments, Inc.*, 52 F.3d 967, 976 (Fed. Cir. 1995) (*en banc*), *aff'd*, 517 U.S. 370 (1996). Claim construction, which serves this purpose, is a matter of law exclusively for the court. *Id.* at 979. The Federal Circuit recently clarified the proper methodology for claim construction in *Phillips v. AWH Corp.*, 415 F.3d 1303 (Fed. Cir. 2005). The court stated that the claims of a patent serve as the proper starting point, noting the “bedrock principle of patent law that the claims of a patent define the invention to which the patentee is entitled the right to exclude.” *Id.* at 1312 (citing *Innova/Pure Water, Inc. v. Safari Water Filtration Sys., Inc.*, 381 F.3d 1111, 1115 (Fed. Cir. 2004) (quotations omitted)). The court articulated that words should generally be given their ordinary and customary meaning – particularly from the vantage point of a person of ordinary skill in the art. *Phillips*, 415 F.3d at 1313. This provides an objective baseline from which claim construction should begin. *Id.*

Significantly, the Federal Circuit further noted that a “person of ordinary skill in the art is deemed to read the claim term not only in the context of the particular claim in which the disputed term appears, but in the context of the entire patent, including the specification.” *Id.* In attempting to discern the meaning of claim terms, the court identified various sources from which the proper meaning may be determined. The claim in which the term appears and other claims of a patent, including both asserted and unasserted claims, can serve as “valuable sources of enlightenment as to the meaning of the claim term.” *Id.* at 1314.

The court also emphasized the primacy of the specification in a claim construction analysis,

noting that it is usually dispositive and “the single best guide to the meaning of a disputed term.” *Id.* at 1315. The specification may reveal whether the patentee acted as his own lexicographer by importing a special definition to the claim term – in which case, the patentee’s lexicography governs. *Id.* at 1316. Moreover, the specification can further reveal any intentional disavowal or disclaimer of claim scope. In such instances, “the inventor has dictated the correct claim scope, and the inventor’s intention, as expressed in the specification, is regarded as dispositive.” *Id.*

The prosecution history should also be taken into consideration if in evidence. Consisting of the complete record of the Patent and Trademark Office (“PTO”) proceedings, “the prosecution history provides evidence of how the PTO and the inventor understood the patent.” *Id.* at 1317. Unlike the specification, however, which represents the final product of ongoing negotiations between the PTO and the patentee, the prosecution history may lack clarity and serve as a less helpful tool in claim construction. *Id.* Nonetheless, this part of the intrinsic evidence “can often inform the meaning of the claim language by demonstrating how the inventor understood the invention and whether the inventor limited the invention in the course of prosecution, making the claim scope narrower than it would otherwise be.” *Id.*

The Federal Circuit cautioned against the use of extrinsic evidence during claim construction since this type of evidence suffers from certain inherent flaws which affect its reliability in a claim construction analysis. This class of evidence includes “all evidence external to the patent and prosecution history, including expert and inventor testimony, dictionaries, and learned treatises.” *Id.* at 1317 (quoting *Markman*, 52 F.3d at 980). Although extrinsic evidence may be useful, “it is unlikely to result in a reliable interpretation of patent claim scope unless considered in the context of the intrinsic evidence.” *Id.* at 1319. A court nonetheless is permitted to admit and use extrinsic

evidence in its sound discretion, so long as the court remains mindful of the inherent flaws in this type of evidence and considers it accordingly. *Id.*

Phillips also clarified the role of dictionaries in claim construction. Placing undue reliance on dictionaries would improperly focus “the inquiry on the abstract meaning of words rather than on the meaning of claim terms within the context of the patent.” *Id.* at 1321. The “ordinary meaning” of the claim term is properly viewed as the “meaning to the ordinary artisan after reading the entire patent.” *Id.* Moreover, dictionaries are naturally suspect as they “provide an expansive array of definitions” and often collect all uses of a word “from the common to the obscure.” *Id.* This may result in extending “patent protection beyond what should properly be afforded by the inventor’s patent.” *Id.* at 1322. Despite such concerns, however, courts are not precluded from using dictionaries in the appropriate manner during claim construction analysis. *Id.*

Lastly, the court must be mindful of the well-settled rule “that while proper claim construction requires an examination of the written description and relevant prosecution history to determine the meaning of claim limitations, additional limitations may not be read into the claims.” *Storage Tech. Corp. v. Cisco Sys., Inc.*, 329 F.3d 823, 831 (Fed. Cir. 2003); *see also In re Donaldson Co.*, 16 F.3d 1189, 1195 (Fed. Cir. 1994) (noting the “general claim construction principle that limitations found only in the specification of a patent or patent application should not be imported or read into a claim.”).

II. The Disputed Claim Terms

A. The '998 Patent

The '998 Patent is entitled "Semiconductor Memory Device With Dual Drivers To Sense Amp Array." The '998 patent was issued on October 1, 1991 to Kannan, *et al.* The '998 Patent primarily concerns the part of the circuitry discussed above that is involved with the amplification of the electrical signals. The invention addresses certain problems that existed in the prior art with respect to the speed and stability of data transmission in the semiconductor memory device. These problems relate to the need to increase the number of memory cells in a DRAM in order to meet the demand for greater memory capacity and higher processing speeds. '998 Patent, col. 1, ll. 9-12. The addition of more memory cells resulted in greater chip size, the use of longer or narrower wires, and increased wiring resistance. *Id.*, col. 1, ll. 13-16.

The first objective of the invention is directed to the arrangement and operation of multiple sense amplifier drivers to counter increased wiring resistance and improve the speed of data transmission. '998 Patent, Abstract & col. 3, ll. 9-12. In the prior art, data in the form of voltages are transferred onto data lines that are connected to a plurality of differential amplifiers. *Id.*, col. 1, ll. 44-46 & Fig. 7. Each differential amplifier has a pair of control terminals – one attached to a restore signal line and the other attached to a drive signal line. *Id.*, col. 1, ll. 46-49. The restore and drive signal lines are connected to a sense amplifier driver which provides the necessary drive signals to activate the differential amplifiers contained within the sense amplifier. *Id.*, col. 1, ll. 55-58. Significantly, the reading operation cannot be completed until all of the differential amplifiers become activated. (Jan. 20, 2006 Certification of Erica S. Helms, Ex. A. at 4:1-2).

The inventors of the '998 Patent recognized that the increased wiring resistances associated

with the restore and drive signal lines caused delay in transmission of the restore and drive signals. More specifically, the signals would have to travel further distances in order to activate all of the differential amplifiers. To solve this problem, the patent teaches the placement of a second sense amplifier driver at the other end of the restore and drive signal lines, i.e., opposite the first amplifier driver. With the additional sense amplifier driver placed in this position, restore and drive signals required to turn on the differential amplifiers can be sent from the second sense amplifier driver in addition to the first. Thus, the signals originating from the first sense amplifier driver need only travel half the distance, thereby reducing the signal transmission time by half. The patent also teaches placing additional sense amplifier drivers within the plurality of differential amplifiers to further decrease wiring resistance and increase the speed of transmission.

In addition, the '998 Patent addresses a second problem that is created by this arrangement. Specifically, the simultaneous operation of a plurality of differential amplifiers causes a peak value of instantaneous current. *Id.*, col. 2, ll. 54-56. This sudden peak in current may result in errors and delay in data transmission. *Id.*, col. 2, ll. 56-64. Thus, a need existed to reduce or eliminate the instantaneous current during operation of the sense amplifier. *Id.*, col. 3, ll. 15-16. The invention sought to correct this problem by incorporating wiring resistances in the signal lines which results in delayed actuation of the second sense amplifier driver. With the differential amplifiers activated at different times, the peak instantaneous current is reduced. *Id.*, col. 4, ll. 30-56.

Based on the parties' written submissions, and as clarified at the *Markman* hearing, there are six disputed claim terms for construction. Claim 1 is a representative claim. It claims:

A semiconductor memory device comprising:

a memory cell array having a plurality of memory cells arranged in the form of a matrix;

a means for reading into a plurality of data lines data from a plurality of memory cells connected to a specific word line by activating said specific word line of said memory cell array;

a sense amplifier consisting of a plurality of differential amplifiers connected to said plurality of data lines respectively for amplifying the data read into said plurality of data lines;

first and second sense amplifier drivers connected to the opposite ends of a restore signal line and a drive signal line respectively which are in turn connected to said plurality of differential amplifiers of said sense amplifiers, and

a delay means for giving a time difference to an operation start timing of said first and second sense amplifier drivers.

'998 Patent, Claim 1. The Court will analyze each disputed claim term in turn.

1. ***“means for reading into a plurality of data lines data from a plurality of memory cells connected to a specific word line by activating said specific word line of said memory cell array”***

The first term appears in Claims 1 and 4 of the patent.⁴ The parties do not dispute that this is a means plus function claim limitation under 35 U.S.C. § 112 ¶ 6. (MEI Patents Chart at 1).

Means-plus-function terms are governed by 35 U.S.C. §112, ¶ 6. This section provides that:

[a]n element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall

⁴ Although the “means for reading” claim term appearing in Claim 7 is not identical to the “means for reading” term appearing in Claims 1 and 4, the parties clarified at the *Markman* hearing that no separate issue existed, and that the Court’s construction for the “means for reading” term in Claims 1 and 4 will apply equally to Claim 7. (Tr. of *Markman* Hr’g at 54:24-55:8).

be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

35 U.S.C. §112, ¶ 6 (Thomson/West 2006). This statutory provision allows patentees to “claim an element of a combination functionally, without reciting structures for performing these functions.” *Apex, Inc. v. Raritan Computer, Inc.*, 325 F.3d 1364,1371 (Fed. Cir. 2003). In exchange for the convenience of employing §112, ¶ 6, the patentee must disclose structure in the specification that is clearly linked or associated with the claimed function. *Budde v. Harley Davidson, Inc.*, 250 F.3d 1369, 1377 (Fed. Cir. 2001).

In construing a means-plus-function claim limitation, the court must engage in a two-part inquiry. *Id.* at 1376. First, the court must identify the particular function that is claimed. *Id.* Second, the court must determine the corresponding structure that performs the claimed function. *Id.* Failure to disclose adequate corresponding structure renders the claim’s scope indefinite, and therefore invalid under § 112, ¶ 2. *Id.* A finding that a claim containing a means-plus-function limitation lacks structural support must be supported by clear and convincing evidence that the specification does not disclose structure sufficient to perform the recited function. *Id.* at 1376-77. Such a determination must be made from the viewpoint of a skilled artisan. *Id.* at 1376.

In addition to agreeing that the disputed term is governed by § 112 ¶ 6, the parties further agree on the claimed function. The stipulated function is “to read data from a plurality of the memory cells connected to a specific word line into a plurality of data lines by activating the specific word line of the memory cell array.” (*Id.*). The dispute arises, however, with respect to the corresponding structure.

The parties’ disagreement stems in part from their varying interpretations of the agreed upon

claimed function. According to Samsung, the claimed function actually encompasses two separate functions, namely “reading” and “activating.” Citing *Asyst Techs., Inc. v. Empak, Inc.*, 268 F.3d 1364 (Fed. Cir. 2001), Samsung argues that the corresponding structure is the structure that is necessary to perform both of these functions. As such, Samsung argues that the corresponding structures are undisclosed structure in the word line driver 3 and the memory array 4. Samsung agrees with MEI that the word line drive 3 performs the “activating” function. (See Samsung Opp’n at 26; Tr. of *Markman* Hr’g at 13:4-6). Samsung, however, claims that some undisclosed structure within memory cell 13 of the memory array 4 is the additional structure required to perform the “reading” function. (Samsung Opp’n at 27).

MEI agrees that the word line driver 3 is corresponding structure, but disagrees that the memory array 4 is necessarily included as structure that performs the claimed function. MEI argues that the claimed function involves only one function, namely “reading . . . by activating,” as opposed to two separate functions as Samsung contends. Relying on the specification, MEI asserts that “reading” is “the result that necessarily flows from the ‘activating’ function,” (MEI’s Br. In Reply to Samsung’s Substituted Opp’n *Markman* Br. for the Matsushita Patents (“MEI Reply”) at 11), which the specification discloses is performed by the word line driver 3. As such, the word line driver is the only corresponding structure.

The Court agrees with Samsung that the claimed function entails the two separate functions of “activating” and “reading.” The Court disagrees with MEI’s position that the “reading” of data should not be construed as a separate function, but instead should be characterized as a necessary result of the “activating” function since, as stated in the specification, data is read “as a result” of a word line becoming activated. Whether it occurs as a result of another function, or is triggered by

some other factor, the act of “reading” data is nonetheless a function. As such, the corresponding structure must include that structure which is necessary to perform both functions. *Micro Chem., Inc. v. Great Plains Chem. Co.*, 194 F.3d 1250, 1257-58 (Fed. Cir. 1999).

The relevant portion of the specification provides a description of the data read-out operation in a semiconductor memory device. Referring to Figure 6 of the patent, the specification states that:

A specific word line in a memory cell array 4 in which a number of memory cells are arranged in the shape of a matrix is *activated* by the output signal of a *word line driver* 3.

As a result, *data from* a plurality of *memory cells* connected to the activated word line *is read out* via a plurality of data lines 5 into a sense amplifier 6.

‘998 Patent, col. 1, ll. 21-28 (emphases added). Based on this excerpt, it is clear that the word line driver and the memory cells which are connected to the activated word line are clearly linked to the data read-out operation. In particular, the word line drive 3 “activates” the word line and the memory cells connected to the activated word line perform the “reading” function.⁵

Samsung argues that the patent fails to describe with specificity what structures within the memory cell actually perform the “reading” function. MEI counters by arguing that the patent need not disclose the structure with specificity because the structure of a memory cell was well known to a person of ordinary skill in the art. This Court agrees. *See S3 Inc. v. NVIDIA Corp.*, 259 F.3d 1364, 1371 (Fed. Cir. 2001) (“The law is clear that patent documents need not include subject matter that

⁵ MEI further argues that the memory cell cannot be corresponding structure because the structure of a memory array is already required by the claim as set forth in the element preceding the means-plus-function limitation. MEI fails, however, to provide support for this argument. (MEI’s Br. at 25; MEI’s Reply at 12). What is clear, however, is that a court must determine all corresponding structure that performs the claimed function when analyzing a means-plus-function limitation under 35 U.S.C. § 112, ¶ 6. *Micro Chem.*, 194 F.3d at 1257-58.

is known in the field of the invention and is in the prior art, for patents are written for persons experienced in the field of the invention.”). Samsung does not present any evidence suggesting that the structure of a memory cell is not commonly understood by persons in the field. Consequently, the Court concludes that the claimed function is “to read data from a plurality of the memory cells connected to a specific word line into a plurality of data lines by activating the specific word line of the memory cell array” and the corresponding structures are the word line driver 3 and memory cells 13.

2. “sense amplifier”

The next disputed term is “sense amplifier” which appears in Claim 1. The claim recites “a sense amplifier consisting of a plurality of differential amplifiers connected to said plurality of data lines respectively for amplifying the data read into said plurality of data lines.” ‘998 Patent, Claim 1. Samsung’s proposed construction limits the “sense amplifier” to include “only differential amplifiers that sense from one memory cell array.” (MEI Patents Chart at 1). Samsung’s position is based on the “consisting of” language which appears in the claim.

Citing *Vehicular Technologies Corp. v. Titan Wheel International, Inc.*, 212 F.3d 1377 (Fed. Cir. 2000), Samsung asserts that “consisting of” is a term of art used in claim construction which signifies restriction and exclusion. Thus, Samsung argues that a sense amplifier is restricted to the claim language following the term “consisting of,” namely “a plurality of differential amplifiers connected to said plurality of data lines respectively for amplifying the data read into said plurality of data lines.” ‘998 Patent, Claim 1. Focusing on the language “said plurality of data lines,” Samsung notes that this term finds its antecedent basis in the preceding element of the claim. As such, it is Samsung’s position that because the claim requires these data lines to “only read from

memory cells on the 'specific word line' in the single memory array, the 'sense amplifier' is likewise restricted to that memory array." (Samsung's Opp'n at 31).

MEI, however, argues that Samsung is impermissibly attempting to incorporate limitations into the plain meaning. MEI contends that the "sense amplifier" is not limited to amplifying data from only one memory cell array. At the *Markman* hearing, MEI clarified that it agrees "consisting of" is a close-ended phrase but posits that the restriction is limited to the "plurality of differential amplifiers" and does not apply to the remaining language following that term. (Tr. of *Markman* Hr'g at 26:2-13).

Based on the claim language, the Court concludes that the "sense amplifier" is not limited to amplifying data from only one memory cell array. In *Vehicular Technologies*, the Federal Circuit stated that:

The phrase "consisting of" is a term of art in patent law signifying restriction and exclusion, while, in contrast, the term "comprising" indicates an open-ended construction. In simple terms, a drafter uses the phrase "consisting of" to mean "I claim what follows and nothing else." A drafter uses the term "comprising" to mean "I claim at least what follows and potentially more."

Vehicular Techs., 212 F.3d at 1382-83 (citations omitted). However, these guidelines provided by the Federal Circuit cannot be neatly applied in this particular instance given the claim language chosen by the patentees. The Court agrees with Samsung that the claim uses "consisting of" with respect to a "plurality of differential amplifiers." The claim, however, proceeds to reference "said plurality of data lines" which finds its antecedent basis in the preceding claim element – notably an element that appears before the "consisting of" language. Instead, this element follows the "comprising" language of the preamble. Thus, the patent claims, at minimum, one memory cell

array, but may potentially claim more, including additional memory cell arrays with additional data lines. The Court finds nothing in the claim language or intrinsic evidence to indicate that the differential amplifiers contained within the sense amplifier are limited to sensing from only one memory cell array. As such, the Court does not find a clear basis for importing the narrowing limitation which Samsung seeks to include. Consequently, the Court adopts MEI's construction.

3. “sense amplifier driver”/“restore signal line”/“drive signal line”

The Court will next construe the claim terms “sense amplifier driver,” “restore signal line,” and “drive signal line.” At the *Markman* hearing, the parties informed the Court that construction of these terms should be addressed together as they present the disputed issue as to whether the claims of the '998 Patent allow for a one-to-one correspondence between the sense amplifier drivers and the differential amplifiers. (Tr. of *Markman* Hr'g at 27:11-36:12, 42:16-19). Based on the parties' arguments, this is an issue that will be dispositive of the patent's validity at a later stage. Though this may be the case, the Court will proceed with its analysis in accordance with Federal Circuit precedent on claim construction, particularly *Phillips*, i.e., by placing primary focus on the intrinsic evidence, and without regard to invalidity arguments disguised as claim construction arguments.

The Court will first address “sense amplifier driver.” Samsung proposes that the term means “a circuit connected to the restore signal line and the drive signal line that provides voltages for operating one or more differential amplifiers.” (MEI Patents Chart at 2). MEI proposes that the term means “a circuit which provides voltages for operating the differential amplifiers. The voltages are provided through the restore signal line and the drive signal line.” (*Id.*). Based on these

constructions, the parties agree that a sense amplifier driver is a circuit that provides voltages for operating differential amplifiers. Thus, the Court will focus its analysis on the two issues arising from the remaining disputed language in the proposed constructions: 1) whether the scope of this term encompasses a sense amplifier driver providing voltage to only one differential amplifier and 2) the proper characterization of restore and drive signal lines with respect to the sense amplifier driver.

The parties have identified the first issue as the “one-to-one correspondence” issue. MEI argues that the sense amplifier driver must be connected to a plurality, i.e., two or more, differential amplifiers, which preclude a one-to-one correspondence between the sense amplifier driver and the differential amplifiers. MEI contends that Samsung’s argument must fail because it inappropriately equates “differential amplifiers” with “sense amplifiers.” (Tr. of *Markman* Hr’g at 28:1-7). In contrast, Samsung offers a number of reasons to support its position that the claim encompasses a one-to-one correspondence. Having reviewed these arguments, the Court finds them unpersuasive and concludes that a circuit encompassing a one-to-one correspondence between a sense amplifier driver and a differential amplifier falls outside of the ambit of the claims of the ‘998 Patent.

The Court will first look at the language of the claims which includes these disputed claim terms. Claim 1 provides, in part, that the semiconductor device is comprised of:

a *sense amplifier* consisting of a plurality of differential amplifiers connected to said plurality of data lines respectively for amplifying the data read into said plurality of data lines;

first and second sense amplifier drivers connected to the opposite ends of a restore signal line and a drive signal line respectively which are in turn connected to said *plurality of differential amplifiers* of said *sense amplifiers*

'998 Patent, Claim 1. Claim 4, another independent claim of the patent, recites in relevant part:

a first sense amplifier consisting of a plurality of differential amplifiers connected to a specific number of data lines of said plurality of data lines respectively for amplifying data read into said specific number of data lines;

a second sense amplifier consisting of a plurality of differential amplifiers connected to remaining data lines of said plurality of data lines respectively for amplifying data read into said remaining data lines;

first and second sense amplifier drivers connected to the opposite ends of the restore signal line and drive signal line which are in turn connected in common to the differential amplifiers of said first and second sense amplifiers;

a third sense amplifier driver connected to said restore signal line and said drive signal line which are disposed between said first and second sense amplifiers

Id., Claim 4. Lastly, Claim 7 which is the last independent claim of the patent requires:

sense amplifiers consisting of a plurality of differential amplifiers for amplifying data read into said plurality of data lines;

a plurality of sense amplifier drivers connected to a plurality of positions of the signal lines respectively which are in turn connected in common to said plurality of differential amplifiers

Id., Claim 7. Based on the claim language, it is clear that the patentees differentiate the terms "sense amplifier" and "differential amplifiers." Notably, in each of these claims, and as discussed above, the patentees chose to describe the sense amplifiers as "consisting of" a plurality of differential amplifiers. Claim 1 only requires one sense amplifier. Claim 4 requires two – i.e., a "first" and a "second" sense amplifier. Claim 7 requires a plurality of sense amplifiers. However, all of these claims share the same requirement that each sense amplifier consists of a plurality of differential amplifiers. Because all of the claims require at least one sense amplifier, the claims also require at

least two differential amplifiers. This eliminates the possibility of a one-to-one correspondence between a sense amplifier driver and a differential amplifier.

The specification and drawings illustrate this point. Figures 1 and 2 depict circuitry in which differential amplifiers 14-18 are contained within the sense amplifier 6. '998 Patent, col. 3, l. 60-col. 4, l. 64. In describing Figure 4, another embodiment of the invention, the specification provides that Figure 4 bears the same reference numbers as Figure 1. *Id.*, col. 5, ll. 14-17. The specification also provides that:

FIG. 4 differs from FIG. 1 in that there are provided two blocks, i.e. first and second sense amplifiers 36 and 37 which correspond to the sense amplifier 6 in FIG. 1, with a third sense amplifier [driver] 38 disposed between said first and second sense amplifiers 36 and 37 to connect it to the latter.

Id., col. 5, ll. 18-23. Thus, the two sense amplifiers 36 and 37 of Figure 4 likewise contain a plurality of differential amplifiers as the sense amplifier 6 of Figure 1.

Samsung fails to provide a sufficient basis to include the proposed limitation of "one of more differential amplifiers" in the construction of the term "sense amplifier driver." First, Samsung argues that its construction relies on the plain meaning of the terms at issue. For the reason discussed above, however, the plain language of the independent claims requires the "sense amplifier" to contain more than one differential amplifier and Samsung fails to convince this Court otherwise.

Samsung also argues that MFI's construction ignores the "comprising" language of the preamble. As previously discussed, this term means that the patentee requires at minimum the elements of the claim, and perhaps even more. Samsung mistakenly argues that this means that the "claim does not exclude additional structure" including a one-to-one "driver to amplifier."

(Samsung's Opp'n at 16). This may be true if by amplifier Samsung is referring to "sense amplifier." But this is not the issue before the Court. Based on Samsung's proposed construction, the issue concerns the one-to-one correspondence between a sense amplifier driver and a differential amplifier which the claim does not encompass. As Samsung acknowledged, the patentees deliberately chose to claim "a sense amplifier consisting of a plurality of differential amplifiers." As such, this eliminates the possibility of a "sense amplifier" containing only one "differential amplifier." Consequently, Samsung's argument on this basis is rejected.⁶

Analysis of the second issue necessarily implicates the proper construction of the disputed terms "restore signal line" and "drive signal line."⁷ Samsung argues that "restore signal line" means "a line having a restore signal connected to one terminal (the restore terminal) of the plurality of differential amplifiers." (MEI Patents Chart at 2). MEI argues that the proper construction is "a line whose purpose is to provide a restore signal to a plurality of differential amplifiers." The parties offer comparable constructions for "drive signal line." (*Id.*).

Samsung asserts that its construction is supported by the plain meaning of the claim language, and in particular the word "connected." Samsung notes that "connected" is the only active verb contained in the claim and contends that the plain meaning of the word means "joined or linked

⁶ Samsung makes alternative arguments concerning the '095 Patent and certain expert testimony. These references are extrinsic evidence. As such, the Court will attribute no weight to these references in light of the clear guidance provided by the intrinsic evidence, namely the claim language and specification.

⁷ The parties stipulated that "restore signal" means "a signal that causes one of the two data lines connected to a differential amplifier to go high when sensing data from a memory cell" and "drive signal" means "a signal that causes one of the two data lines connected to a differential amplifier to go low when sensing data from a memory cell." (MEI Patents Chart at 2).

together.” (Samsung Opp’n at 15). According to Samsung, the claim does not include any verbs describing the “purpose” of the line and MEI provides no basis for including such limitation in the claim construction. Samsung argues that Federal Circuit case law prohibits importing the purported “purpose” of the signal lines in the claim construction.

MEI counters by arguing that its construction does take into account the term “connected” by acknowledging that voltages are provided from the sense amplifier driver to the differential amplifiers through the restore and drive line signals which are connected to both. MEI further argues that Samsung’s use of the term “having” in its construction is vague and therefore should be rejected.

The Court, however, finds that the claim language and specification provide a clear meaning for the terms “restore signal line” and “drive signal line.” The relevant claim language provides that sense amplifier drivers are “connected to the opposite ends of a restore signal line and a drive signal line respectively which are in turn connected to said plurality of differential amplifiers of said sense amplifiers.” ‘998 Patent, Claim 1. Thus, based on the claim language alone, it is clear that the restore and drive signal lines are physically connected to both the sense amplifier drivers and to the plurality of differential amplifiers contained within the sense amplifiers.

The Court next looks to the specification. The specification describes the restore signal line and the drive signal line as follows:

First, when data from a memory cell 13 is sensed by data lines 5-1-5-5, there will arise slight potential differences between the signal lines of each pair of data lines 5-1-5-5. Then, drive signals are applied to the input terminals 23 and 24 of the first sense amplifier driver 7 so as to make the transistors 21 and 22 of the sense amplifier driver 7 conduct. Consequently, *electric current flows in the restore signal line 19 and the drive signal line 20*, and thus, *the differential amplifiers 14-18 start to operate*. The operation as described above is same as in the prior art.

'998 Patent, col. 4, ll. 14-24. The specification teaches that in addition to being connected to the sense amplifier drivers and the differential amplifiers, electric current flows through the restore signal line and drive signal line which causes the differential amplifiers to begin operation. Thus, in light of the plain language of the claim and the specification, the Court concludes that "restore signal line" means "a line that is connected to both the sense amplifier driver and the differential amplifier through which an electric current, namely a restore signal, flows which causes the differential amplifiers to operate." The Court finds that this construction adequately describes the "restore signal line" and finds no basis for importing the specific limitation of "purpose" – a term that is absent in both the claim language and specification. (Tr. of *Markman* Hr'g at 49:14-18). Similarly, "drive signal line" means "a line that is connected to both the sense amplifier driver and the differential amplifier through which an electric current, namely a drive signal, flows which causes the differential amplifiers to operate."

Returning to the first issue presented with respect to the construction of "sense amplifier driver," the Court concludes that MEI's proposed definition more closely reflects the Court's conclusion regarding the restore and drive signal lines. Thus, the Court will adopt a slightly modified construction. Accordingly, "sense amplifier driver" means "a circuit that is connected to a restore signal line and a drive signal line which provides voltages for operating the differential amplifiers."

4. "opposite ends"

The next dispute concerns the claim term "opposite ends" which appears in Claims 1 and 4 of the patent. Claim 1 requires the first and second sense amplifier drivers to be connected to the "opposite ends" of a restore signal line and a drive signal line. '998 Patent, Claim 1. MEI proposes

that the sense amplifier drivers are on “opposite ends” of the restore and drive signal lines “with respect to groupings of differential amplifiers.” (MEI Br. at 30). Relying on the specification, MEI argues that the patent teaches a reduction in wiring resistances when sense amplifier drivers provide signals from opposite ends of a grouping of differential amplifiers, and not necessarily the opposite ends of the restore and drive signal lines. As such, “opposite ends” should be construed as opposite “with respect to a plurality of differential amplifiers.” (*Id.* at 32).

Samsung counters by arguing that the term “opposite ends” unambiguously references the restore and drive signal lines, and not the differential amplifiers. Thus, Samsung offers the proposed construction for “opposite ends” as “towards the termination points of conductor [i.e. restore/drive signal] lines.” (Samsung Opp’n at 32). Samsung contends that its construction is based on the “heavy presumption” that the term should be construed based on its ordinary meaning and the plain language of the claim. Samsung also maintains that the intrinsic evidence supports this construction.

Based on the claim language, the Court agrees with Samsung that the term “opposite ends” refers to the “opposite ends” of the restore and drive signal lines and does not refer to the grouping of differential amplifiers. Mindful of the bedrock principal that “the claims of a patent define the invention to which the patentee is entitled the right to exclude,” *Phillips*, 415 F.3d at 1312 (citing *Innova/Pure Water*, 381 F.3d at 1115), a review of the ‘998 Patent’s independent claims leads to this conclusion. The Court first notes that the term “opposite ends” is used in an identical manner in Claims 1 and 4. Notably, Claim 7 does not reference the term “opposite ends.” In both Claims 1 and 4, the claim requires a first and second sense amplifier driver to be located at opposite ends of the signal lines. Claim 4 specifically requires:

first and second sense amplifier drivers connected to the opposite ends of the restore signal line and drive signal line which are in turn connected in common to the differential amplifiers of said *first and second sense amplifiers*

Id., Claim 4. As discussed extensively above, each sense amplifier consists of a plurality of differential amplifiers. As such, in this particular claim there are at least two separate groups of differential amplifiers belonging to a first and a second sense amplifier. Moreover, Claim 4 requires a third sense amplifier driver to be located at some point between the first and second sense amplifiers. Specifically, the claim requires:

a third sense amplifier driver connected to said restore signal line and said drive signal line which are disposed between said first and second sense amplifiers

Id., Claim 4. The Court concludes that this language belies MEI's argument that the "opposite ends" are in reference to the opposite ends of a group of differential amplifiers. If MEI's construction were applied to Claim 4, it would be unclear as to whether the first and second sense amplifier drivers were positioned at the opposite ends of the first group of differential amplifiers, i.e., those belonging to the first sense amplifier, or the second group of differential amplifiers belonging to the second sense amplifier – since this claim necessarily includes a third sense amplifier driver separating these two groups. The more reasonable construction is that the first and second sense amplifier drivers in both Claims 1 and 4 are positioned at the opposite ends of the restore and drive signal lines, i.e., towards the termination points of the signal lines, which would allow the third sense amplifier driver to be positioned in between the two.

The language of Claim 7 further supports the Court's construction. Although Claim 7 contains similar limitations as those found in Claims 1 and 4, the term "opposite ends of a restore

signal line and a drive signal line" is not present. If MEI's construction were correct, one would expect the claim to retain the term "opposite ends" even without the accompanying phrase "restore signal line" and "drive signal line" since the claim refers to a plurality of differential amplifiers. But it does not which further supports the conclusion that the "opposite ends" in Claims 1 and 4 refer to the ends of the restore and drive signal lines.

The Court also rejects MEI's assertion that the specification supports its construction. First, MEI points to the specification which reads:

FIG. 1 is different from FIG. 6 only in that the sense amplifier 6 has one end connected to a first sense amplifier driver 7, and has its other end connected to a second amplifier driver 29.

A p-type field effect transistor 30 and an n-type field effect transistor 31 are connected to the other end of the sense amplifier 6, i.e. the other end of a restore signal line 19 and the other end of a drive signal line 20 respectively, said transistors 30 and 31 constituting a second sense amplifier driver 29.

Id., col. 3, ll. 61-64, col. 4, ll. 1-7. However, this part of the specification is describing an embodiment with one grouping of differential amplifiers positioned in between a first and second sense amplifier driver. Thus, although the specification describes sense amplifier drivers positioned at the opposite ends of a plurality of differential amplifiers, the specification also describes them equally positioned at opposite ends of the restore and drive signal lines. Therefore, the Court finds no basis in this disclosure of the patent to conclude that the "opposite ends" means with respect to the group of differential amplifiers, and not to the signal lines.

Second, MEI refers to the specification describing Figures 4 and 5 of the Patent.

Such an arrangement ensures that the opposite ends of the first sense amplifier 36 will be driven by means of the first and second sense amplifier drivers 7 and 38 and that the second sense amplifier 37 will

be driven by means of the third and second sense amplifier drivers 38 and 29.

Id., col. 5, ll. 34-39. The Court disagrees with MEI that this language supports its construction. Instead, this language shows that the patentees used the term “opposite ends” consistently in the patent. In this instance, however, “opposite ends” is used to modify the “first sense amplifier 36,” as opposed to a “restore signal line and a drive signal line” as Claims 1 and 4 require. Specifically, referring to Figure 5, the patentees state that the opposite ends of the first sense amplifier 36 will receive drive signals from sense amplifier driver 7 and 38, which according to Figure 5 are located toward the termination points of sense amplifier 36. Consequently, having rejected MEI’s proposed construction, the Court adopts Samsung’s construction that “opposite ends” means “toward the termination point of the conductor lines.”

B. The ‘095 Patent

The ‘095 Patent is entitled “Semiconductor Memory Apparatus With Reduced Line Widths.” The patent was issued on December 20, 1994 to Yamada *et al.* The ‘095 Patent seeks to address problems that existed in the prior art as the need for greater memory and high speed operation for DRAM increased. As discussed above, increasing the memory capacity in DRAM required the integration of additional memory cells. To maintain chip size, however, the width of connecting lines in the chip needed to be reduced. A consequence of using narrower wires in a DRAM was greater wiring resistance. This led to the delay in amplification of electrical signals from the bit line and the overall delay in the operational speed of the DRAM chip.

The inventors of the ‘095 Patent attempted to solve this problem by reorganizing the spatial

arrangement of the DRAM chip components. Rather than positioning sense amplifier drive circuits along the periphery of a memory cell array, the patent teaches distributing the sense amplifier drive circuits throughout a region containing memory cells and sense amplifier circuits. The inventors recognized that “a sense amplifier circuit that is located close to the switches that connect the bit line pair to the power supply voltages will exhibit a relatively low amount of amplification delay.” ‘095 Patent, col. 1, ll. 45-48. The patent further teaches using power supply meshes with the memory array region to provide sufficient drive current for operation. This arrangement places the sense amplifier drive circuits closer to the power supply voltages. The invention also eliminates the need for long thin wires to provide power to the sense amplifier drive circuits. See ‘095 Patent, col. 8, ll. 26-28, col. 10, ll. 17-24, col. 12, ll. 49-57, col. 14, ll. 50-62.

1. “*distributed*”

The ‘095 Patent presents two main disputed claim construction issues. The first concerns the arrangement of the elements of Claim 1 based on the claim terms “distributed through” and “distributed among.” Independent Claim 1 recites:

A semiconductor memory apparatus including a memory array region having formed therein:

a plurality of unit memory cell blocks *distributed through* said memory array region at regular spacings, each formed of an array of memory cells;

a plurality of unit sense amplifier blocks *distributed among* said unit memory cell blocks at regular spacings, each formed of an array of sense amplifier circuits;

a plurality of sense amplifier drive circuits for driving said sense amplifier circuits, *distributed among* said unit sense amplifier blocks at regular spacings; and

first and second voltage supply meshes, mutually electrically isolated and each extending throughout said memory array region, respectively coupled to receive first and second supply voltages;

each of said sense amplifier drive circuits being coupled to an adjacent point on said first voltage supply mesh to receive said first supply voltage and to an adjacent point on said second voltage supply mesh to receive said second supply voltage.

'095 Patent, Claim 1. MEI argues that Claim 1 requires the first three elements of the claim, namely the unit memory cell blocks, the unit sense amplifier blocks, and the sense amplifier drive circuits, to be arranged "throughout the power grid in two dimensions of the memory array region." (MEI Br. at 42). Samsung, however, contends there is no requirement that the distribution must be in two dimensions. Instead, the distribution must be "in at least one direction." (MEI Patents Chart at 10).

A review of the intrinsic evidence supports Samsung's construction. Beginning its analysis with the claim language, the Court notes that the plain language of Claim 1 does not include any language suggesting that the unit memory cell blocks, the unit sense amplifier blocks, and the sense amplifier drive circuits must be arranged in a two-dimensional distribution. Rather, the claim requires the three elements to be distributed "at regular spacings" throughout the memory array region. The Court agrees with Samsung that this can be achieved with a one-dimensional arrangement.

MEI argues that the fourth element of the claim concerning the first and second voltage supply meshes indicate that two-dimensions are required. MEI asserts that because the sense amplifier drive circuits must be "coupled to an adjacent point" on the first and second voltages supply meshes, which the parties agree are a two-dimensional arrangement of conductors, (MEI Patents Chart at 11), the sense amplifier drive circuits must also be in a two-dimensional

arrangement. (MEI's Br. at 42). The Court disagrees that the claim language warrants such a conclusion. Although the claim may require two-dimensional power supply meshes, the sense amplifier drive circuits may nonetheless be coupled to adjacent points of the meshes if they were arranged in a one-dimensional distribution.

Samsung argues that the patentees in Claim 6 claimed a spatial arrangement in two directions, i.e., along columns and rows, which establishes that a two-dimensional arrangement was not claimed in Claim 1. Claim 6 reads:

said plurality of unit memory cell blocks are configured as a *plurality of columns* of unit memory cell blocks each extending in a *first direction*, each column consisting of a *plurality of said unit memory cell blocks disposed at regular spacings*

'095 Patent, Claim 6. MEI counters by arguing that Claim 6 covers other elements in addition to the two-dimensional arrangement, such as word lines formed in an upper and lower layer, and therefore cannot be a basis for concluding that the patentees did not claim a two-directional arrangement in Claim 1.

The Court finds Samsung's argument to be more persuasive. Although the claim may include additional elements, Claim 6 expressly requires the unit memory cell blocks to be arranged in a plurality of columns, i.e., two or more columns. The claim further specifies that each column contains two or more unit memory cell blocks. This establishes an arrangement in two dimensions – specifically, columns of memory cells blocks distributed at regular spacings in one direction, and multiple memory cell blocks distributed within each column at regular spacings in a second direction. As the Federal Circuit noted in *Phillips*, “the presence of a dependent claim that adds a particular limitation gives rise to a presumption that the limitation in question is not present in the

independent claim.” *Phillips*, 415 F.3d at 1315. The Court finds this to be the case here.

Turning to the specification, MEI argues that the purpose of the invention was to improve the performance of memory arrays by placing “sense amplifier drive circuits closer to the mesh source of the power, so that the resistances of the lines providing power to the sense amplifier drive circuits are substantially reduced.” (MEI Br. at 43). MEI claims that “[o]nly the grid arrangement of sense amplifier drive circuits connected at adjacent points to the meshes would accomplish this purpose.” (*Id.*). MEI posits that sense amplifier blocks and unit memory cells must also be arranged in two dimensions to achieve the purpose of the invention.

The Court disagrees with MEI’s position. The specification describes the invention as an improvement over the prior art by distributing sense amplifier drive circuits throughout a memory array region, rather than only at the periphery as they were in the prior art. In describing the problems in the prior art, the patent discloses that:

The basic cause of delay in . . . amplification by the sense amplifier circuits is the time required to charge the bit line pair capacitance, through pairs of sense amplifier drive lines (each coupled in common to a large number of sense amplifier circuits) that are peripherally connected to power supply voltages after a data read operation is initiated. The rate of charging of the bit line pair capacitance of a sense amplifier circuit depends upon the position of the sense amplifier circuit along such a pair of sense amplifier drive lines, i.e. *a sense amplifier circuit that is located close to the switches that connect the bit line pair to the power supply voltages will exhibit a relatively low amount of amplification delay, whereas a sense amplifier circuit that is located substantially distant from these switches will exhibit a substantial amount of amplification delay.*

’095 Patent, col. 1, ll. 35-51. Thus, the patent teaches that a sense amplifier circuit’s position in relation to the power supply voltages (the voltages in this case being V_{ss} and V_{dd}) largely dictates the amplification delay. The patent further teaches that using power supply meshes to provide voltages

to sense amplifier drive circuits that are distributed throughout the memory array region reduces the delay. *Id.*, col. 1, l. 63-col. 2, l. 12. The specification does not state that delay may be reduced only if the various circuitry are configured in a two-dimensional arrangement, as MEI contends.

The patent specification further discloses that the power supply meshes and the sense amplifier drive circuits are dispersed throughout the memory array region, as opposed to being located at the periphery. *Id.*, col. 8, ll. 17-25. This allows each of the sense amplifier drive circuits to be connected to the nearest points on the power supply meshes to receive the V_{ss} and V_{dd} supply voltages. *Id.* The specification further provides that this arrangement reduces the path length between each sense amplifier drive circuit and the power supply meshes, as well as the distance between the sense amplifier circuits and the sense amplifier drive circuits. *Id.*, col. 8, ll. 26-33. Moreover, the specification discloses that the sense amplifier drive circuits drive “a smaller number of sense amplifier circuits than is possible in a prior art semiconductor memory apparatus, in which sense amplifier drive circuits are not distributed throughout the memory cell array.” *Id.*, col. 8, ll. 33-38.

The Court concludes that all of these inventive features may be accomplished by an arrangement that extends in one direction, such as the illustration shown in Figure 5B. Although MEI accurately notes that this figure is merely a “cut-out of Figure 5A,” (Tr. of *Markman* Hr'g at 79:13), it nonetheless depicts an arrangement that incorporates two power supply meshes in two dimensions, as well as a plurality of sense amplifier drive circuits 103 dispersed throughout the memory array region rather than along the periphery of the memory array, a plurality of unit memory

cell blocks 102, and sense amplifier circuits 108 which all extend in one direction.⁸ Notably in this configuration, the length of the paths between the sense amplifier drive circuit and the power source of the V_{ss} and V_{dd} voltages, as well as the path between the sense amplifier circuits and the sense amplifier drive circuits, are reduced in comparison to the prior art DRAM arrangement as depicted in Figure 1A. As such, the Court finds no basis in the specification that warrants inclusion of the “two-dimension” limitation in the claim construction.

The prosecution history further supports the Court’s construction. MEI claims that the arrangement of a “two-dimensional power grid with an arrangement of unit memory cell blocks each having adjacent unit sense amplifier blocks and sense amplifier drive circuits distributed through the grid” was repeatedly emphasized as one of the novel features during prosecution. (MEI Br. at 44). The Court finds, however, that the patentees emphasized the novelty of the two-dimensional power supply mesh and its role in allowing each sense amplifier circuit to be connected to a power supply voltage source at a position adjacent to the sense amplifier circuit during prosecution. Specifically, the patentees argued:

It is clear that [the problem of varying amounts of sensing delay time for the bit line pairs] results from the fact that it is necessary in the prior art to connect each drive line pair to (effectively) the supply voltage source at the periphery of the chip. However with the present invention that limitation is removed. By forming a 2-dimensional low-impedance power supply mesh that extends throughout the sense

⁸ MEI urges the Court to focus on Figure 5A, rather than on the incomplete illustration of Figure 5B, which purportedly demonstrates that the claim requires two-dimensional distribution. (Tr. of *Markman* Hr’g at 79:13). Although the figure does indeed illustrate a two-dimensional arrangement, the Court notes that Figure 5A merely represents one embodiment of the invention. ‘095 Patent, col. 3, ll. 1-3. It is well-established that “claims are construed in light of the specification, and are not limited to a designated ‘preferred embodiment’ unless that embodiment is in fact the entire invention presented by the patentee.” *Vulcan Eng’g Co., Inc. v. Pata Aluminum, Inc.*, 278 F.3d 1366, 1376 (Fed. Cir. 2002).

amplifier region of the integrated circuit, it is arranged that each sense amplifier circuit can be connected (through respective switches) to that supply mesh at a position adjacent to the sense amplifier circuit. Thus, each sense amplifier circuit can be directly connected via a low-impedance path to the power supply voltage source, and so the respective rates of charging of the various bit line pairs can be made substantially identical.

(Aug. 30, 1993 Amendment at 3). Notably, the patentees did not emphasize that the sense amplifier circuits, memory cell blocks or sense amplifier drivers must be configured in a two-dimensional arrangement as well. Rather, the focus remained on the low-impedance path provided by power supply meshes to the power supply voltage source.

The patentees continued to exalt the benefits provided by the power supply meshes by providing an example concerning a prior art chip and a chip embodying the present invention.

It must be emphasized that the impedance of one of the drive lines 7, 8 in the prior art circuit of Fig. 1A, as measured between any specific position within the integrated circuit and the periphery of the chip, will be very much greater than the impedance of the power supply grid, measured between a similar position within an integrated circuit of the present invention and the periphery of the chip. That will be true even if the length and cross-sectional area of each conductor constituting a prior art drive line are identical to those of each conductor constituting the supply voltage mesh of the present invention, and is due to the fact that the vertical and horizontal conductors of the mesh are periodically interconnected (by through-hole connections) as shown in the drawings. That is the essential basis of the invention.

(*Id.* at 3-4). Essentially, the patentees compared structurally equivalent chips, one representing the prior art and the other representing the claimed invention, and described the difference in the degree of impedance in the drive signal lines between identical positions within the integrated circuit and the periphery of the chip. Noting that the impedance would be much lower in the claimed invention, the patentees attributed the lower impedance to the presence of a two-dimensional power supply

mesh and not to any two-dimensional arrangement of the sense amplifier circuits, sense amplifier driver circuits, and memory cell blocks.

Lastly, the Examiner withdrew final rejections of the claims after having concluded that they were patentable over the Tobita reference because Tobita lacked the two-dimensional power supply mesh. (Sept. 29, 1993 Examiner's Action at 2). The Examiner, however, proceeded to reject the claims on 35 U.S.C. § 103 grounds in view of the Kohno and Tobita references. In a request for reconsideration, the patentees argued that these references failed to suggest or disclose the claimed voltage supply meshes. The patentees asserted that the basis of the invention was "the combination of a V_{ss} power supply mesh and a V_{dd} power supply mesh, which are interrelated such as to enable a distributed arrangement of sense amplifiers that are connected within the power supply meshes, whereby enhanced speed of operation is achieved for the sense amplifiers" (Apr. 4, 1994 Request for Reconsideration at 3-4). Based on these statements, and the aforementioned statements made by the patentees during prosecution of the patent, it is clear that the patentees identified the power supply meshes as the main inventive features of the patent, and not a two-dimensional arrangement of the first three elements appearing in Claim 1. Consequently, the Court will adopt Samsung's proposed construction.

2. *"upper layer of a substrate"/"lower layer of said substrate"*

The next disputed claim terms appear in asserted Claim 2 of the '095 Patent. Claim 2, which depends from Claim 1, recites a semiconductor memory apparatus:

in which each of said [] first and second voltage supply meshes is formed of a plurality of first supply lines extending in a first direction and formed in *an upper layer of a substrate* of said semiconductor memory apparatus and a plurality of second supply lines extending in a second direction and formed in *a lower layer of said substrate* which is electrically isolated from *said upper layer*, and a plurality of

through-hole connections for interconnecting said first and second supply lines at points of intersection thereof.

'095 Patent, Claim 2. The parties disagree on the construction of "an upper layer of a substrate" and "a lower layer of said substrate." There are two disputed issues presented by these claim terms. The first issue concerns the proper construction for "upper layer" and "lower layer." (MEI Patents Chart at 12). The second issue is whether the upper and lower layers are actually part of the substrate itself, or if they are positioned on a substrate. (Tr. of *Markman* Hr'g at 88:18-23, 91:19-21).

MEI argues that the upper and lower layers are on the substrate. MEI points to the specification, particularly Figure 5E, to support its position. MEI asserts that accepting Samsung's definition would be error because it reads out this preferred embodiment of the claim. (*Id.* at 90: 7-15). Additionally, MEI relies on expert testimony for the proposition that one of ordinary skill in the art would not place conductors "in" the base substrate because such an arrangement would not work well from a practical standpoint. (Certification of David L. Taylor ("Taylor Cert.") ¶ 14).

In contrast, Samsung argues that the layers are part of the substrate. Focusing on the plain language of the claim, Samsung asserts that the ordinary meaning of the term "of" must apply which is "used as a function word to indicate the whole that indicates the part denoted by the preceding word <most [of] the army>." WEBSTER'S 9TH NEW COLLEGIATE DICTIONARY 819 (1991). According to this ordinary meaning, Samsung contends that the substrate (the whole) includes the layer (the part). Samsung further contends that MEI's reliance on Mr. Taylor's comments are not relevant to claim construction and MEI's construction finds no support in the original application of the patent.

The Court disagrees with Samsung that the ordinary meaning of the term "of" clearly defines the disputed terms. The Federal Circuit emphasized in *Phillips*, that "the ordinary and customary

meaning of a claim term is the meaning that the term would have to a person of ordinary skill in the art in question at the time of the invention” *Phillips*, 415 F.3d at 1313. In light of this principle, and post-*Phillips* Federal Circuit case law applying the clarified methodology, the Court must exercise caution when relying on the ordinary meaning of claim terms. See *Curtiss-Wright Flow Control Corp. v. Velan, Inc.*, No. 05-1373, 2006 WL 335609, at * 4 (Fed. Cir. Feb. 15, 2006) (concluding that the district court placed “too much emphasis on the ordinary meaning of the term ‘adjustable’ without adequate grounding of that term within the context of the specification”). Thus, rather than readily accepting Samsung’s definition for the singular term “of,” a generic term that undoubtedly may be defined in a number of ways, the Court must determine how a skilled artisan would understand the terms “upper layer of a substrate” and “lower layer of said substrate” in the context of the specification.

The Court notes that reference to the term “substrate” appears in three areas of the written description of the patent. The first appears in column 2, lines 35 to 44. This section essentially mirrors the language of claim 2 and therefore does not provide much information beyond what one finds in the claim itself. The second is the portion of the specification describing Figure 5E.⁹ This

⁹ Citing *TurboCare Division of Demag Delaval Turbomachinery Corp. v. General Electric Co.*, 264 F.3d 1111, 1118 (Fed. Cir. 2001), Samsung argues that Figure 5E was not part of the original disclosure, and thus cannot be considered by the Court during its claim construction analysis. (Tr. of *Markman* Hr’g at 93:9-14). In *TurboCare*, the Federal Circuit upheld the district court’s ruling that a claim was invalid because it failed to satisfy the written description requirement. In reaching this conclusion, the court engaged in an extensive analysis as to whether the type of spring and the location of that spring constituted new matter. As to the type of spring, the court concluded that a genuine issue of material fact existed. As to the location of the spring, the court concluded that this failed to satisfy the written description requirement. Here, to the extent Samsung is arguing that a claim or claims of the ‘095 Patent is invalid because it fails to satisfy the written description requirement, Samsung fails to provide any supporting argument or facts to warrant such a finding. Instead, Samsung merely states in a conclusory manner that Figure 5E and the accompanying language were not part of the original

drawing illustrates a cross-sectional view of an embodiment of the invention. ('095 Patent, col. 8, ll. 51-61). The specification provides that the lower layer is "a polysilicon layer formed on substrate." (*Id.*, col. 8, ll. 53-55). The lower word lines 3' and lower lines of the supply meshes 11b and 12b, which are composed of suitably doped polysilicon, are formed within this layer. (*Id.*). In addition, "[v]arious other circuit elements such as transistors are also formed in that polysilicon layer." (*Id.* col. 8, ll. 55-57). Upper word lines 3" and the upper lines of the supply meshes 11a and 12a are formed of metal and formed in the upper layer. (*Id.*, col. 8, ll. 57-59). The patent discloses that this layer is formed above the lower polysilicon layer and is insulated from it. (*Id.*, col. 8, l. 59-61).

The Court finds that this portion of the specification provides clear guidance with respect to the first issue concerning the construction for "upper layer" and "lower layer." Noting that the patent describes the "upper layer" as being formed "above . . . the polysilicon lower layer," MEI's proposed definition for "upper layer," namely "a layer above a lower layer," accurately reflects this description more than Samsung's construction and will therefore be adopted. Similarly, "a lower layer" will be construed as "a layer below an upper layer."

This patent disclosure, however, does not provide the same clear guidance as to the second issue – particularly in light of the last portion of the specification which references the substrate. This is found in columns 11 and 12. The patentees describe in detail the circuitry of the first embodiment of the invention and the read-out operation sequence. In particular, the specification notes that an n-channel MOS transistor is used in the sense amplifier driver circuit 57 to drive the

disclosure of the patent and therefore cannot be considered by the Court. (*Id.*). Samsung's argument is inapt and is therefore rejected.

sense amplifier via the drive line. The patent further discusses an advantage of using the n-channel MOS transistor.

Moreover it is advantageous to use an n-channel MOS transistor as the transistor 62 in each sense amplifier drive circuit 57, since in general, for a specific size of transistor (i.e. *size of substrate area occupied*), an n-channel MOS transistor has a greater current drive capability than a p-channel MOS transistor. Thus, savings in space are achieved by using n-channel MOS transistors.

'095 Patent, col. 12, ll. 4-11. Thus, in referencing a transistor, which has already been described as being located within the lower layer, (*id.* col. 8, ll. 55-57), the patent equates the size of a transistor to the size of the substrate area which it occupies. This suggests that patentees considered the "upper" and "lower" layers containing the word lines, the supply mesh lines and other circuit elements, such as transistors, to be part of the substrate.

In line with the teachings of *Phillips*, the Court concludes that the "upper" and "lower" layers are part of the substrate. This construction is based on the ordinary meaning of the word "of" when read in the context of the specification – specifically, the disclosure in the patent which teaches that other circuit components, such as transistors, occupy the substrate area. *See Phillips*, 415 F.3d at 1321 (noting that reliance on a dictionary to determine the meaning of a word within the context of the patent, rather than in the abstract, is appropriate). Moreover, that Court finds nothing in the specification indicating that a substrate cannot be composed of multiple layers.

The Court disagrees with MEI's assertion that adopting this construction would read out Figure 5E, a preferred embodiment, from the claim. As Figure 5E represents a cross-sectional view of a part of a semiconductor memory apparatus, there is no indication that the substrate cannot encompass the areas above and around the upper and lower layers, such that the layers would be part

of the entire substrate.¹⁰ Therefore, MFI's claim construction argument is rejected.

Consequently, the Court will construe the terms in the following manner which is consistent with this conclusion: 1) "an upper layer of a substrate" means "a layer, situated above a lower layer, which is part of the substrate; 2) "a lower layer of said substrate" means "a layer, situated below an upper layer, which is part of the substrate."

C. The '648 Patent

The '648 Patent is entitled "Redundancy Semiconductor Memory Device Which Utilizes Spare Memory Cells From A Plurality of Different Memory Blocks, And Utilizes The Same Decode Lines For Both The Primary And Spare Memory Cells." This patent issued on December 12, 1995 to Atsushi Fujiwara. The '648 Patent is directed toward an improved redundancy technique in which a defective primary memory cell is replaced by a spare memory cell in a semiconductor memory device. In the semiconductor industry, a redundancy scheme is important since semiconductor memory devices would normally be discarded in its entirety if the device contained even one defect. Employing a technique that allows for correction of a defect in the device leads to a reduction in manufacturing costs.

The invention of the '648 Patent allows a defective memory cell in one memory cell block to be corrected not only by a spare memory cell within the same memory cell block, but by a spare

¹⁰ Moreover, the Court does not find Mr. Taylor's statements particularly helpful. Mr. Taylor states that conductive layers would not be placed in the substrate because the chip would not work properly and this would be contrary to customs in the field. Mr. Taylor, however, only provides a conclusory explanation as to why this is so. (Taylor Cert. ¶ 14; Supp. Taylor Cert. ¶ 19). As such, the Court will attribute no weight to these statements. *See Phillips*, 415 F.3d at 1321 (stating that "conclusory, unsupported assertions by experts as to the definition of a claim term are not useful to a court").

memory cell located in another memory cell block. Thus, “even in cases where a defective memory cell occurs in one memory cell array that has no spare memory cells left, a spare memory cell in the other memory cell array is available.” ‘648 Patent, col. 3, ll. 29-32. Additionally, the patent employs the concept of “shared decode lines” in which “the spare memory cell is connected to the primary memory cell in the same memory cell block with a shared decode line and is driven through a shared word line drive signal generation circuit.” *Id.* col. 3, l. 67-col. 4, l. 3. This allows for the elimination of separate circuitry for the primary and spare word lines, which reduces the overall size of the memory chip. *Id.* col. 4, ll. 3-8.

1. “*memory replacement means*”

This first disputed term concerning the ‘648 Patent is “memory replacement means” which is found in Claims 1 and 4 of the patent. The parties agree that this term is a means plus function limitation. In Claim 1, the parties stipulate that the function of the “memory replacement means” is “to store in advance an address of a defective primary memory cell and, when an address of the memory cell to be accessed matches the address of the defective memory cell, to cause the decoder to select a spare memory cell to replace the defective primary memory cell.”¹¹ (MEI Patents Chart at 17). The parties agree that Embodiment IV discloses one alternative for corresponding structure,

¹¹ Claim 4 involves a slightly different function for “memory replacement means.” The parties agree that the claimed function is “to control the decoder such that when an address of the memory cell to be accessed matches the address of the defective primary memory cell, the decoder selects a spare memory cell to replace the defective primary memory cell. The selected spare memory cell may be selected from a memory cell block other than the memory cell block that contains the defective primary memory cell.” (MEI Patents Chart at 21). The parties disagree on the corresponding structure. The parties informed the Court that the arguments made with respect to the corresponding structure of claim 1 equally apply to claim 4. (Tr. of *Markman* Hr’g at 97:14-19). As such, the Court’s construction for the corresponding structure of Claim 1 will also be the corresponding structure for Claim 4.

namely the redundancy-use decision circuit 20, and the decode signal switching circuit 19 in Figure 8. (*Id.*). The parties dispute, however, what corresponding structures, if any, are disclosed in the other three embodiments described in the patent.

a. Embodiment I and III

The parties propose that Embodiments I and III of the patent disclose one alternative of the corresponding structure. The parties agree that the corresponding structure includes the redundancy-use decision circuit 20, the redundancy encoder 8, and the decode signal selection circuit 9 in Figures 1 and 7, but disagree as to whether the row predecoder 5 is included as corresponding structure.

MEI argues that the specification clearly discloses the row predecoder 5 as structure that performs the claimed function. MEI asserts that the OR gates, which are part of the decode signal selection circuitry 9, must receive two input signals in order to produce an output signal which ultimately enables the row decoder 4 to select a spare row. (Taylor Cert. ¶ 24). One input is the redundancy decode signal 11 which originates from the redundancy encoder 8. The second is the signal generated by the row predecoder 5. (*Id.*). Once the OR gate receives both signals, it then produces a binary output signal that is sent to the row decoder 4 for spare row selection. (*Id.* ¶¶ 24-25). Thus, MEI contends that if the decode signal selection circuit 9 does not receive signals from both the row predecoder and the redundancy encoder, a malfunction may result and a spare row would not be selected. (*Id.* ¶¶ 25-27; Tr. of *Markman* Hr'g at 102:17-23).

In response, Samsung argues that the row predecoder 5 should not be included as corresponding structure because it does not perform the function of selecting the spare word line. Instead, Samsung contends that the specification indicates that the row predecoder 5 only selects the primary word line. Samsung points to the part of the specification which reads:

Row predecoders 5l and 5r are provided, which send out predecode signals 10l (for the block group SA1) and 10r (for the block group SA2) respectively *to select a primary memory cell* corresponding to the address of address signal input.

'648 Patent, col. 7, ll. 1-5 (emphasis added). Citing *Asyst Technologies, Inc. v. Empak, Inc.*, 268 F.3d 1364 (Fed. Cir. 2001), Samsung argues that structure that merely enables other structure to perform the claimed function does not qualify as corresponding structure. Samsung contends that by admitting that the decision signal selection circuit 9 would not work if inputs from the row predecoder 5 were eliminated, MEI essentially admits that the row predecoder 5 is merely an enabler, and not corresponding structure, in accordance with *Asyst*.

The Court finds that MEI has the more persuasive argument. The claimed function consists of two parts: 1) to store in advance an address of a defective primary memory cell and 2) to cause the decoder to select a spare memory cell. It is undisputed that the redundancy use decision circuit 20 corresponds to the first part of the function. See '648 Patent, col. 1, l. 61-col. 2, l. 1 ("Such a redundant configuration contains . . . a redundancy-use decision circuit 20, comprised of elements SD0 to SDn to store the addresses of defective memory cells and to decide whether an input address corresponds to the stored address . . ."). Thus, the Court must determine what structure "cause[s] the decoder to select a spare memory cell."

In the detailed description of Embodiment I, the specification describes the operation of a semiconductor memory device in accordance with the invention. Notably, it provides:

As a means for replacing a defective memory cell with a spare memory cell, the redundancy-use decision circuits 20l and 20r, the row predecoder 5l and 5r, and the decode signal selection circuits 9l and 9r are provided. This allows the row decoders 4a to 4d of the memory cell blocks a to d to operate efficiently, and a word line corresponding to the address of address signal can be driven without increasing the memory chip area.

'648 Patent, col. 10, ll. 32-39. The first sentence indicates that the row predecoder 5 structure is involved with the process of replacing a defective memory cell. Samsung argues, however, that this is not so because the last sentence of this excerpt indicates that the function being discussed is "to operate efficiently." The Court disagrees. When read in context, particularly with the full disclosure describing Embodiment I, the function being described is "replacing a defective memory cell with a spare." The Court does not read "to operate efficiently" as a separate and distinct function, but rather a characterization of the claimed function.¹²

Notably, in Column 9 of the specification, the patent describes the process in which a spare word line is selected. The specification describes how the decode signal selection circuit 9 takes the input signals from the row predecoder and the redundancy decode signals to yield two output signals that become a "1." The specification proceeds to describe how the row decoder 4 receive these signals, and ultimately a spare word line is selected. In particular, in describing Figures 3 and 4, the specification provides:

The input of the predecoders A to C of the row predecoder 5r becomes a "0." Of redundancy decode signal 11r only signals RA0 and RB0 are supplied at a "1." Accordingly, in the decode signal selection circuit 9r, only predecode signal WP0 and first gate voltage application signal XPA0 become a "1." In the row decoder 4r, shown in FIG. 4, a voltage is applied to the element WD0 of the word line drive signal generating circuit 21, and through the AND gates only the spare word lines become a "1", as a whole. From among the spare word lines, *a spare word line SWa, which corresponds to the element WD0, is selected.*

¹² In addition to this argument, Samsung further contends that this specification fails to establish that the row predecoder 5 is corresponding structure because the redundancy encoder 8, which the parties have agreed is corresponding structure, is not mentioned in this excerpt. The Court disagrees. As explained below, this excerpt cannot be read in isolation, but rather in the entire context of the specification which supports the conclusion that the row predecoder 5 is corresponding structure. Therefore, Samsung's argument is rejected.

'648 Patent, col. 9, ll. 6-17 (emphases added). MEI's expert explains that "each of the OR gates in the decode signal selection circuit 9 in Embodiment 1 of the '648 Patent . . . requires both an input from the row predecoder 5 and an input from the redundancy decode signal 11 to generate an output which is to be sent to the row decoder 4 to enable its selection of a spare row." (Taylor Cert. ¶ 27). Without input from the row predecoder, MEI asserts that the OR gate would not work.

Samsung fails to rebut this proposition. In Samsung's opposition brief, Samsung states that the specification makes clear that "the inputs from the row predecoder 5 are essentially ignored by the decode signal selection circuit 9 on a spare access, which instead selects the inputs from the redundancy encoder 11." (Samsung Opp'n at 51 (citing '648 Patent, col. 7, ll. 1-18, and particularly 15-18)). The part of the specification relied upon by Samsung, however, fails to address what takes place in the decision signal selection circuit 9 itself as it selects either the primary or spare word line. Therefore, based on the specification, and the clarification provided by the expert opinion of Mr. Taylor, the Court accepts MEI's assertion that the row predecoder 5 is corresponding structure that performs the function of "caus[ing] the decoder to select a spare memory cell."¹³

The Court also disagrees with Samsung that this conclusion violates Federal Circuit precedent set forth in *Asyst*. In *Asyst*, the Federal Circuit concluded that a communication line (line

¹³ At the hearing, Samsung also stated that its expert testified "that when a defective word line is being addressed, the row predecoder is effectively disabled." (Tr. of *Markman* Hr'g at 110:10-13). Samsung failed to provide the Court with any citation to the expert report in its brief and at the hearing. This alone is a sufficient basis to reject this argument. However, having reviewed the report on its own, the Court nonetheless concludes that Samsung's expert fails to explain how the row predecoder is not necessary structure. Samsung's expert merely states that "the row predecoder 5 is not part of the corresponding structure because it does not perform the function – it is irrelevant to the selection of a spare memory," but does not address the operation that takes place in the decision signal selection circuit. (Gosney Report ¶ 100). As such, the Court accepts MEI's proposition.

51) was not corresponding structure that performed the recited function of “receiving and processing digital information transmitted between the second microcomputer means and the tool-mounted communication means.” (*Asyst*, 268 F.3d at 1370). The court noted that the communication line was “not referred to at any point in the description of [the] functions of local control processors 20. Instead, line 51 [was] simply identified as the communication line between the communicating means 50 and the process controller 20.” *Id.* This differs from the present case. Here, the row predecoder 5 is referenced in the specification as discussed above. Moreover, the claimed function here is broader than the functions of “receiving and processing information” which were the functions at issue in *Asyst*. The Court finds that the specification discloses the row predecoder 5 as part of the corresponding structure which “causes” the decoder to select a spare word line. Accordingly, the Court concludes that the row predecoder 5, along with the redundancy-use decision circuit 20, the redundancy encoder 8, and the decode signal selection circuit 9 in Figures 1 and 7, are corresponding structures for the “memory replacement means.”

b. Embodiment II

The parties present a second disputed issue with respect to the “memory replacement means” claim term. The issue is whether Embodiment II is inoperative. Samsung argues that Embodiment II is inoperative for two reasons. First, Samsung contends that the embodiment lacks a structure that enables it to select between primary and spare signals, such as the decision signal selection circuit 9 found in Embodiments I and III. Second, Samsung asserts that this embodiment uses only one primary memory stop signal that effectively disables the circuit by turning off all four of the decoders. (Tr. of *Markman* Hr’g at 120:24-121:5).

MEI responds by arguing that a decision signal selection circuit is not necessary in this

particular embodiment to perform the claimed function, and that the specification discloses the structures that perform the function. This includes the redundancy-use decision circuit 20, the redundancy encoder 8, the circuit for generating the primary memory stop signal 18, and the memory cell block selection switching circuit 29 in Figure 6. (MEI Patents Chart at 17). Additionally, MEI argues that Samsung provides no basis for its position that all of the row decoders 4 are turned off when they receive the same primary memory stop signal 18.

The Court agrees with MEI and finds that Embodiment II is not inoperative for the reasons set forth by Samsung. Rather than determining whether Embodiment II discloses a structure that is analogous to the decision signal selection circuit, a structure described in other embodiments in the patent, the proper inquiry before the Court is whether the specification describing this particular embodiment discloses corresponding structure that performs the claimed function of “stor[ing] in advance an address of a defective primary memory cell and, when an address of the memory cell to be accessed matches the address of the defective memory cell, [] caus[ing] the decoder to select a spare memory cell to replace the defective primary memory cell.” *See Kemco Sales, Inc. v. Control Papers Co., Inc.*, 208 F.3d 1352, 1360 (Fed. Cir. 2000) (“Once a court establishes that a means-plus-function limitation is at issue, it must construe that limitation, thereby determining what the claimed function is and what structures disclosed in the written description correspond to the ‘means’ for performing that function.”). The Court finds that it does.

The relevant portion of the specification describing the function of “storing” defective primary memory cell addresses and “caus[ing]” the decoder to select a primary memory cell is found in Columns 11 and 12 of the patent. The specification describes the operation for selecting a spare word line in this particular embodiment. In the event a spare word line must be used, a redundancy

use signal is first sent by the decision circuit 20. '648 Patent, col. 11:67-12:17. Two other signals, the redundancy decode signal 11 and redundancy block selection signal 31 are then sent by the redundancy encoder 8. Yet another signal, the primary memory stop signal 18, is then fed to the row decoders 4 to prevent the primary word lines from operating. As this occurs, the block selection switching circuit 29 translates a signal into a redundancy block selection signal 31. Since no primary word lines can be selected, the spare word line is selected. *Id.*

The Court finds that the specification identifies sufficient corresponding structure to perform the claimed function, particularly "to cause the decoder to select a spare memory cell." The mere absence of a decision signal selection circuit 9 or some analogous structure does not alter this conclusion. Moreover, as MEI correctly notes, the specification discloses that the primary word lines are disabled after receiving the primary memory stop signal 18. It does not disclose that all of the row decoders are disabled, nor does it suggest that spare word lines are disabled. Consequently, the Court concludes that Embodiment II is operative and the corresponding structure is the redundancy-use decision circuit 20, the redundancy encoder 8, the circuit for generating the primary memory stop signal 18, and the memory cell block selection switching circuit 29 in Figure 6.¹⁴

2. Claim 2

The last dispute regarding the '648 Patent concerns Claim 2. Claim 2, which depends from Claim 1, recites:

A semiconductor memory device as in claim 1, wherein the memory replacement means includes:

a redundancy-use decision circuit that stores in advance the address

¹⁴ In the event the Court concludes that Embodiment II is operative, the parties agree that these structures are corresponding structures. (MEI Patents Chart at 17).

of said defective memory cell and outputs a redundancy-use signal if the address signal agrees with the stored address;

a predecoder responsive to the address signal and sends out a predecode signal to select a word line of the primary memory cell corresponding to the address signal; and

a decode signal switching circuit that connects the predecode signal, supplied from the predecoder, to the word line drive signal generating circuit, while on the other hand connecting, when said redundancy-use signal is sent out from the redundancy-use decision circuit, the redundancy-use signal instead of the predecode signal to the word line drive signal generating circuit.

'648 Patent, Claim 2. In their claim construction briefs, the parties appeared to identify the disputed issue as whether Claim 2 should be construed as a means plus function claim governed by § 112 ¶ 6. The parties informed the Court at the *Markman* hearing, however, that this was not a proper characterization of the disputed issue. (Tr. of *Markman* Hr'g at 128:10-15). Rather, the parties identified the issue as "whether [one] could ever infringe claim 2, a dependent claim, without infringing Claim 1, which is a means-plus-function." (*Id.* at 128:18-21). Put differently, the issue concerned the scope of Claim 2. (*Id.* at 131:20-22).

At the hearing, Samsung stated that it understood MFI's position to be that Claim 2 could be broader than Claim 1. (*See id.* at 130:23-131:1 ("The argument, as I've understood it from them, is that this claim could actually be broader than Claim 1 because you could infringe it without having a redundancy encoder and that's what we object to.")). In response, MFI explained its position and ultimately stated that Samsung's understanding was incorrect. (*See id.* at 132:17-18 ("So Samsung's premise that this is a broader claim is incorrect.")). The parties proceeded to inform the Court:

Mr. Sherwood: Your Honor, what I understood [Ms. Etienne] to just say is that Claim 2 is narrower in scope than Claim 1, which is our whole point.

The Court: Is that correct, narrower in scope than Claim 1.

Ms. Etienne: If you infringe Claim 2, you will infringe Claim 1, yes, your Honor.

The Court: All right. Thank you.

(*Id.* at 132:25-133:8). Based on this colloquy, it appears that the parties ultimately resolved the issue on their own. Since the issue concerning Claim 2 no longer appears to be in dispute, the Court refrains from addressing this issue any further.

D. The '048 Patent

The '048 Patent is entitled "Semiconductor Device Having a Contact Window Structure." The '048 patent was issued on May 12, 1998 to Lee *et al.* The patent discloses a structure of a semiconductor memory device having self-aligned contact windows, which permit electrical contact between the source and drain regions of a semiconductor device.

In the prior art, the conventional manufacturing process for a semiconductor device involved the formation of a gate insulating layer, a polycrystalline layer, and an oxide layer over a semiconductor substrate. After the basic layers are in place, an etching process occurs which removes exposed portions of the layers to form the gate structures. '048 Patent, col. 1, ll. 56-67. Source and drain regions are formed on the substrate, and an additional etching process takes place. After this process, a contact window is formed and a conductive layer is added. Should the oxide layer become over-etched, exposure of the conductive layer of the gate may result – which ultimately may lead to electrical shorts.

The '048 Patent solves this problem by claiming a semiconductor device with a self-aligned contact window that is formed by "the sequential formation of a gate insulating layer, a

polycrystalline silicon layer, a tungsten silicide layer, and a first insulating layer on a semiconductor substrate.” ‘048 Patent, col. 2, ll. 48-51. The layers are then sequentially dry etched, “thereby forming the gates out of said first and second conductive layers and, at the same time, opening a contact window to an associated source and drain.” ‘048 Patent, col. 2, ll. 53-56. By using a process wherein the tungsten silicide layer is etched considerably faster than the other layers, an undercut region of the layer is formed which protects the corners of the gates from exposure. As such, the reliability of the device is increased.

1. ***“recessed edge portion”/“edge portion”***

The first issue regarding the ‘048 Patent concerns the terms “recessed edge portion” and “edge portion” which appear in Claim 1 of the patent. Claim 1, the only independent claim of the ‘048 Patent, reads:

A semiconductor device with a gate structure, said semiconductor device comprising:

a gate insulating layer formed on a semiconductor substrate;

a gate structure having a first conductive layer of polycrystalline silicon and a second conductive layer of refractory metal silicide, said first conductive layer being formed on said gate insulating layer and said second conductive layer being formed on said first conductive layer, said second conductive layer having a *recessed edge portion* with respect to an *edge portion* of said first conductive layer; and

a first insulating layer formed on a top surface of said second conductive layer and having a protrusion portion with respect to an edge of said second conductive layer.

‘048 Patent, Claim 1. Based on the parties’ arguments and proffered constructions, the disputed issue is whether the claim requires the “edge portion” to be recessed in its entirety or if only a part of the “edge portion” need be recessed. To answer this question, the Court must first construe “edge

portion.”

Samsung defines “edge portion” as “one of the narrow surfaces of the layer.” (Samsung Patent Chart at 3). MEI contends that “edge portion” is properly construed as a “part of the edge.” (*Id.*). Samsung argues that its construction is consistent with basic grammar principles. Specifically, Samsung asserts that “portion” is a noun that refers to a portion of the associated layer, and the terms “recessed” and “edge” are adjectives that describe the “portion.” MEI argues that the claim language, as well as the purpose and description of the invention support its construction that “edge portion” only refers to a part of an edge.

The Court agrees with Samsung’s construction. The Court concludes that the claim language and the parties’ stipulated constructions provide a clear construction for this term. The Court notes that the parties agree that “edge” is defined as the “two-dimensional border or boundary” such as a “sidewall” of a layer. (Samsung Patent Chart at 3). The parties further agree that the term “portion” means “a part of a whole.” (*Id.*). The claim recites “said second conductive layer having a *recessed edge portion* with respect to an *edge portion* of said first conductive layer.” ‘048 Patent, Claim 1 (emphases added). Based on the plain language of the claim, the Court finds that the term “portion” clearly refers to a part of the associated layer. At the *Markman* hearing, the parties agreed that “edge portion” and “recessed edge portion” should be construed similarly, except that the latter is recessed. (Tr. of *Markman* Hr’g at 147:18-148:3). Thus, “recessed edge portion” refers to a part of the second conductive layer that is recessed and “edge portion” refers to a part of the first conductive layer.

The claim further specified that the “portion” of the associated layer to which the claim is referring is the “edge” of the layer. Based on the stipulated construction and a plain reading of the claim language, the term “edge portion” is therefore best construed as “the two-dimensional border

or boundary of the layer.” The Court rejects Samsung’s proposed construction which incorporates the limitations “narrow surface” since the Court cannot find, nor does Samsung identify, a proper basis for importing this additional language. Additionally, the Court rejects MEI’s construction that the term should be defined as “part of the edge.” As stated above, the “whole” to which the term “portion” is referring is the layer, and not the edge. The Court further disagrees with MEI’s assertion that by including the word “portion,” the terms “edge” and “edge portion” cannot possibly have the same meaning. This presumes that the term “edge” is being used as a noun, rather than a term describing the “portion” of the layer. As discussed above, a plain reading of the claim indicates that the “portion” is the edge of the layer.

Having construed the critical term “edge portion,” the Court turns its focus on the term “recessed edge portion.” Samsung argues that the entire edge of the second conductive layer must be indented. Samsung relies on the plain language of the claim, the specification, and the cited art during prosecution. MEI contends that only part of the edge must be recessed. More specifically, MEI explains that the claim would cover a structure “where the recess of the sidewall is not throughout its height but involves at least a recess of the critical edge portion of the sidewall of the second conductive layer.” (MEI’s Br. In Supp. of Its Claim Construction of the Samsung Patent (“MEI Opp’n”) at 13). MEI asserts that its construction is supported by a fair reading of the claim language and the purpose of the invention.

Based on the intrinsic evidence, the Court concludes that the term “recessed edge portion” requires the entire edge of the layer to be indented in its entirety. The Court first looks to the claim itself. Having construed the term “edge portion” to mean the entire sidewall of a layer, the Court finds no language in the claim itself indicating that the patentees intended for only part of the edge

to be recessed.

This comports with the written description and the figures of the patent. Figures 2B to 2E, which illustrate the process for manufacturing a semiconductor memory device in accordance with the invention, each depict a recessed edge portion of the second conductive layer that is fully recessed with respect to the first conductive layer. Moreover, the description in the specification accompanying these drawings suggest that the "recessed edge portion" or the "undercut" is recessed in its entirety. For example, as the patent describes the etching process, the specification discloses that "[t]he tungsten silicide layer 34 is etched faster than the first insulating layer 35, forming an undercut 38 under the first insulating layer 35." '048 Patent, col. 3, ll. 39-41. The specification further states that "a treatment of an SCl solution for 10-60 minutes is applied to enlarge the undercut 38." *Id.*, col. 3, ll. 49-50. In further describing the process, the specification provides:

Therefore, it is seen that the tungsten silicide layer 34 is not externally exposed in the contact window, even though the corner of the first insulating layer 35 is slightly etched when etching the second insulating layer 39, because the tungsten silicide layer 34 has the undercut 38 portion under the first insulating layer 35.

Id., col. 4, ll. 12-17. Thus, as the patent describes the undercut being formed by the application of particular etching processes, the specification gives every indication that these processes affect the exposed portions of layer, or the sidewalls, indiscriminately which leads to a fully recessed edge portion. The specification provides no indication that the etching process only causes part of the exposed portions of the layer to become recessed. As such, the Court finds no basis in the intrinsic evidence for concluding that the claim requires only part of the edge portion to be recessed, as

opposed to the entire edge portion.¹⁵

2. “formed on”

The next disputed term “formed on” appears in Claim 1 of the patent. The claim requires various layers to be “formed on” the substrate and other layers. For example, Claim 1 requires “a gate insulating layer *formed on* a semiconductor substrate.” ‘048 Patent, Claim 1 (emphasis added). The claim further requires a first conductive layer to be “formed on” the gate insulating layer. Samsung proposes that the term means “above and supported by *or* in contact with.” (Samsung Patent Chart at 1) (emphasis added). MEI asserts that the term means “directly over *and* in contact with.” (*Id.*). Based on a comparison of these proposed constructions and the parties’ arguments, the disputed issue before this Court is whether the claim term requires direct contact between the layer and the element following the term “formed on.”

The Court first looks to the claim language. Claim 1 uses the term “formed on” to describe the placement of the layers in the semiconductor memory device. First it recites that a gate insulating layer is “formed on” a semiconductor substrate. Second, a first conductive layer is “formed on” the gate insulating layer. Next, the second conductive layer is “formed on” a first conductive layer. And last, a first insulating layer is “formed on a top surface” of a second conductive layer. A plain reading of this language suggests that these layers are formed directly on and in contact with the associated layer since each layer is described in sequence. Had the patentees intended to allow additional layers positioned in between these four layers, the claim could have recited that the first

¹⁵ MEI also argues that Samsung’s construction ignores the real world of semiconductor processing because recessed sidewalls are rarely perfectly vertical. This argument, however, is based on the expert opinion of Mr. Coburn. This qualifies as inherently suspect extrinsic evidence to which the Court attributes little to no weight during its analysis, particularly in light of the clear guidance provided by the intrinsic evidence.

conductive layer, the second conductive layer, and the first insulating layer were all “formed on” a gate insulating layer. Instead, the patentees chose to draft the claim in a way that places emphasis on the positioning of each individual layer thereby suggesting that the layers are in contact with the associated layer (or the semiconductor substrate in the case of the gate insulating layer).¹⁶

Samsung argues that the patent’s specification establishes that direct contact is not required.

Samsung refers to the part of specification which provides:

According to the present invention, a self-aligned contact window is formed by the *sequential formation* of a gate insulating layer, a polycrystalline silicon layer, a tungsten silicide layer, and a first insulating layer on a semiconductor substrate.

‘048 Patent, col. 2, ll. 47-51 (emphasis added). The Court finds, however, that this disclosure supports a construction requiring contact because it uses term “sequential formation,” indicating that the layers are formed one on top of the other.

Figures 2A to 2E, as well as the description of the preferred embodiments, further support this construction. These figures illustrate the process for manufacturing the invention. The figures depict each layer being in contact with the associated substrate or layer, providing no suggestion that other layers can be interspersed within these layers. Moreover, the patent’s description of the etching process which creates the desired the layers’ recessed and protruded edge portions further suggests that these layers must be in contact with one another. The specification provides that the “etch rates of the first insulating layer 35, the tungsten silicide layer 34, and the polycrystalline silicide layer 33

¹⁶ Samsung makes this argument with respect to the Abstract of the patent. (Samsung Br. at 22). However, it is well-settled that the claims of the patent are what defines the invention, not the written description. *See Phillips*, 415 F.3d at 1312 (noting the “bedrock principle of patent law that the claims of a patent define the invention to which the patentee is entitled the right to exclude.”).

have a ratio of 30:140:24, respectively.” ‘048 Patent, col. 3, 55-57. The specification then provides:

Therefore, it is seen that the tungsten silicide layer 34 is not externally exposed in the contact window, even though the corner of the first insulating layer 35 is slightly etched when etching the second insulating layer 39, because the tungsten silicide layer 34 has the undercut 38 portion under the first insulating layer 35. Therefore, a safe spacing between the gates 37 is maintained in the formation of the self-aligned contact window 42.

Id., col. 4, ll. 12-19. Based on this disclosure, the patent teaches that a self-aligned contact window, a main inventive feature of the patent, results from a particular etching process which etches the tungsten silicide layer considerably faster than the other two layers. This teaches away from the concept of adding additional layers in between these three layers since it could lead to different etching ratios. Thus, the Court finds that the specification likewise supports a construction that requires contact between the layers.

The Court, however, finds the prosecution history to be most informative of the patentees’ intended meaning for the phrase “formed on.” Claim 1, as originally submitted, stated that:

A semiconductor device with a gate structure, said semiconductor device comprising . . . a first insulating layer *formed on* said second conductive layer and having a protrusion portion with respect to an edge of said second conductive layer.

(Nov. 3, 1995 Prelim. Amendment at 1-2). The Examiner originally rejected the claim stating that it was obvious in light of the Hsu and Takeuchi references. The Examiner noted:

Hsu et al shows in fig. (10) a semiconductor device with a gate structure, the semiconductor device comprising: a gate insulating layer (30) formed on a semiconductor substrate; a gate structure 32, 34) having a first conductive layer (32) of polycrystalline silicon and a second conductive layer (34) a refractory metal, the first conductive layer (32) being formed on the gate insulating layer and the second conductive layer (34) being formed on the first conductive layer (32),

the second conductive layer (34) having a recessed edge portion with respect to an edge portion of the first conductive layer; and a first insulating layer (50) formed on the second conductive layer (34) and having a protruding portion with respect to an edge of the second conductive layer (the first insulating layer protrudes from the edge of the second conductive layer), but omits the formation of the silicide as in the claimed invention.

(Feb. 16, 1996 Examiner's Action at 3). Based on Figure 10 of the Hsu patent, the Examiner clearly used the phrase "formed on" to mean "in contact with." Notably, Figure 10 illustrates the first insulating layer being in direct contact with the sides of the second conductive layer (34), and not above the layer.

In response to this rejection, the patentees amended the claim to specify that a first insulating layer is "formed on *a top surface of*" a second conductive layer. (May 28, 1996 Amendment at 3, 5) (emphasis added). Samsung argues that this amendment supports its construction because if the patentees intended "formed on" to mean "in contact with," this amendment would have been unnecessary. This Court disagrees. Based on the rejection, it is clear that the patentees did not add the phrase "a top surface" to clarify *whether* the first insulating layer is in contact with the second conductive layer, since Figure 10 of the Hsu patent illustrates that the layers were in direct contact, but to distinguish *where* the point of contact existed. Thus, by adding the phrase "a top surface of," the patentees used the term "formed on" consistent with the Examiner's use, and clarified that the first insulating layer of the invention was on a top surface of a second conductive layer, rather than at the sides of the second conductive layer as depicted in the Hsu patent. As such, the Court finds Samsung's argument unavailing and concludes that the prosecution history supports a construction

requiring direct contact.¹⁷

Therefore, based on the intrinsic evidence, the Court concludes that the claim term "formed on" requires direct contact between the layer and the element following the term in the claim.¹⁸

3. "layer"

The next disputed term is "layer" which appears throughout all of the claims of the patent. The disputed issue is whether the term "layer" may be composed of multiple layers. Samsung argues that it can be, proffering the construction of "a thickness of material laid on, spread over, or otherwise covering a surface; a stratum." (Samsung Patent Chart at 1). In support of its position, Samsung refers to the specification which describes the gate insulating layer as being made of successively stacked sublayers. '048 Patent, col. 3, ll. 18-20. Samsung further relies on how the term "layer" is used in other art that was cited by the patentees during prosecution.¹⁹

In contrast, MEI argues that "layer" means "a thickness of a single material. A single

¹⁷ Samsung also argues that certain prior art references which were cited by the patentees during prosecution, particularly the Okumura and Satoh patents, support its construction because the drawings accompanying the patents' use of the term "formed on" does not show any actual contact. Though this may be the case, the Court attributes more weight to the patentees' amendment since it was made to clarify the term "formed on," thereby providing greater insight into the patentees' intended meaning.

¹⁸ Samsung also relies on extrinsic evidence, namely a dictionary definition for the term "on," to support its construction. However, as MEI correctly notes, this definition is particularly unreliable given the additional definitions provided in the same dictionary and a subsequent edition that equally supports MEI's construction. (MEI Opp'n at 20-21). Thus, the Court finds this dictionary definition insufficient to alter the Court's conclusion which is based on the intrinsic evidence.

¹⁹ Samsung further argues that MEI use the term "layer" in a manner consistent with its proposed construction in its own patent applications. As this material is clearly extrinsic evidence divorced from the '048 Patent, the Court will not consider these references in its analysis.

material means a material of homogenous composition.” (Samsung Patent Chart at 1). MEI asserts that this construction is consistent with the claim language because it describes a sequential formation of different layers. Additionally, MEI argues that the specification also describes the sequential formation of layers. (MEI Opp’n at 24 (citing ‘048 Patent, col. 1, ll. 47-56; col. 2, ll. 15-26)). MEI contends that the layers could be made up of an element (e.g. polysilicide) or a compound (e.g. tungsten silicide).

Having reviewed the parties’ arguments, the Court concludes the parties present two separate issues with respect to the claim term “layer.” The first issue is determining the construction of the generic term “layer.” The second issue is whether the particular term “gate insulating layer” can consist of multiple layers.²⁰

With respect to the term “layer,” the parties agree that a layer is “a thickness of material.” MEI seeks to import the limitation that the layer consist of a single of homogenous material. The Court, however, does not find a proper basis to import this limitation in the claim construction. The term “homogenous” does not appear in the claim language. Moreover, the portions of the specification which MEI references describe different types of materials that can be used to form the layers. For example, the specification indicates that:

The gate insulating layer 32 is formed of an oxide, a silicon nitride, or successively stacked layers of an oxide and a nitride.

‘048 Patent, col. 3, ll. 18-20. The Court disagrees with MEI that this language forms a sufficient

²⁰ MEI asserts that the parties are not disputing the term “gate insulating layer.” (Tr. of *Markman* Hr’g at 158:10-14). Based on the parties’ arguments, however, the Court disagrees. Both parties are relying on the part of the specification describing the “gate insulating layer” in support of their construction for the term “layer.” Having reviewed the arguments and the intrinsic evidence, the Court concludes that construing these terms individually would provide the greatest resolution to the parties dispute.

basis to import the “homogenous” limitation since it describes only one of the preferred embodiments. *See Bell Atl. Network Servs., Inc. v. Covad Commc’n Group, Inc.*, 262 F.3d 1258, 1273 (Fed. Cir. 2001) (noting “it is generally impermissible to limit claim terms by a preferred embodiment or inferences drawn from the description of a preferred embodiment”). As such, the Court will adopt Samsung’s definition based on the ordinary meaning of “layer.” Consequently, “layer” means “a thickness of material laid on, spread over, or otherwise covering a surface; a stratum.”

MEI argues that Samsung’s definition is flawed because it fails to distinguish between one layer (singular) and multiple layers (plural). This leads to the next disputed issue concerning the “gate insulating layer.” Based on the same portion of the specification referenced above, the patentees made clear that the “gate insulating layer” may be composed of multiple layers, or sublayers. Consequently, in the context of the patent, the “gate-insulating layer” is one particular layer that may consist of multiple layers.

4. “gate structure having a first conductive layer of polycrystalline silicon and a second conductive later of refractory metal silicide”

The issue presented with this claim term is whether the gate structure is limited to these two particular layers or whether it may include additional layers. Samsung contends that the gate structure may consist of additional layers. Samsung asserts that the term “having” is used as an open-ended phrase in various parts of the claim and the specification. Samsung further notes that the claim itself is devoid of any language to suggest that it should be construed as close-ended phrase.

MEI argues that “having” does not carry any presumption that the term is open-ended and asserts that the claim is limited to a two-layer gate structure consisting of a layer of polycrystalline

silicon and a layer of a refractory metal silicide. Relying on the plain reading of the claim, the specification, expert opinion, and art cited during prosecution, MEI argues that the claim requires a very specific gate structure consisting of these two layers.

The Federal Circuit discussed the effect of the term “having” when used in a claim. In *Crystal Semiconductor Corp. v. TriTech Microelectronics International, Inc.*, 246 F.3d 1336, 1348 (Fed. Cir. 2001), the court stated:

When a patent claim uses the word “comprising” as its transitional phrase, the use of “comprising” creates a presumption that the body of the claim is open. In the parlance of patent law, the transition “comprising” creates a presumption that the recited elements are only a part of the device, that the claim does not exclude additional, unrecited elements.

The transition “having” can also make a claim open. However, the term “having” does not convey the open-ended meaning as strongly as “comprising.” “Having,” for instance, does not create a presumption that the body of the claim is open. Therefore, this court examines the claim in its full context to determine whether Crystal’s use of “having” limits claim 1 to its recited elements.

Id. at 1348 (citations omitted). Thus, the Court must examine the full context of the claim and the specification to determine whether the term “having” is so limited.

Turning first to the claim language, the claim requires: “a gate structure having a first conductive layer of polycrystalline silicon and a second conductive layer of refractory metal silicide.” Thus, the plain language of the claim appears to limit the gate structure to the first and second conductive layers. The Court rejects Samsung’s argument that the term “having” must be construed as open-ended because the patentees used the term in an open-ended manner in other parts of the claim and in the specification. As the Federal Circuit intimated in *Crystal Semiconductor*, the focus is not on the term “having” in isolation, but on the patentees’ use of the term “having” with respect

to the "gate structure." As such, the Court must look to the full context of the patent to determine whether "having" limits the gate structure to the recited elements of the first and second conductive layer.

The written description suggests the same conclusion. In particular, the "Summary of the Invention" states:

According to the present invention, a self-aligned contact window is formed by the sequential formation of a gate insulating layer, a polycrystalline silicon layer, a tungsten silicide layer, and a first insulating layer on a semiconductor substrate. Then, the gate insulating layer, the polycrystalline silicon layer, the tungsten silicide layer, and the first insulating layer are sequentially dry etched, *thereby forming the gates out of said first and second conductive layers* and, at the same time, opening a contact window to an associated source and drain.

'048 Patent, col. 2, ll. 47-56. Thus, the written description shows the intent to make the gate structures out of the first and second conductive layers. Moreover, Figure 2B of the preferred embodiment illustrates that gates 37 are formed by the polycrystalline silicon layer 33 and the tungsten silicide layer 34.

The prosecution history also describes the gate structure in the same manner. During prosecution, the patentees described the problems associated with the reduced reliability of the contact in conventional devices. The patentees explained that the invention of the '048 Patent solved this problem, in part, "by providing a semiconductor device including a gate structure in which a second conductive layer of refractory metal silicide is formed on a first conductive layer." (May 28, 1996 Amendment at 4).

Having examined the claim in the full context of the patent, the Court concludes that the patent drafters used the term "having" as a close-ended phrase. Consequently, the "gate structure

having . . .” term means that the gate structure consists of a first conductive layer of polycrystalline silicon and a second conductive layer of refractory metal silicide only, and may not include additional layers.

5. *“refractory metal”*

The last dispute involves the term “refractory metal.” Samsung proposed the term should be construed as “a metal, generally having characteristics including a high melting point, low electrical resistivity, corrosion resistance, and high temperature stability.” (Br. In Supp. of Samsung’s Proposed Claim Construction for the ‘048 Patent Counterclaim at 36). MEI asserts that the proper construction is “a metal having a high melting point in excess of 2000° C.” (Samsung Patent Chart at 1). Based on these constructions and the parties’ arguments, the issue is whether refractory metal includes titanium.

Samsung argues that it does. Samsung argues that a person of ordinary skill in the art would consider a variety of properties such as those listed in its proposed construction to determine whether a metal is a refractory metal. Additionally, Samsung contends that the art cited during prosecution of the ‘048 clearly includes titanium as a refractory metal, despite the fact that the melting point of titanium is below 2000° C. Moreover, Samsung argues that MEI has characterized titanium as a refractory metal in other patents it owns.

MEI arrives at its proposed construction by listing five metals that a person of ordinary skill in the art would regard as a refractory metal. MEI then focuses on the melting points of these metals which range from 2477° C to 3422° C. In an effort to provide a clear demarcation for refractory metals, MEI chose a cut-off melting point temperature of 2000° C. MEI relies on expert opinion in formulating this construction, and contends that it is also supported by numerous technical sources.

The Court must look to the intrinsic evidence to determine the best interpretation of this term. The claim itself does not provide much guidance in defining the term, as it only refers to "refractory metal." The written description indicates that tungsten is construed as a refractory metal. See '048 Patent, col. 3, ll. 20-23 ("A polycrystalline silicon layer 33 and a refractory metal silicide layer 34 of, for example, tungsten silicide are then sequentially formed on the gate insulating layer 32 via an oxidation, CVD or PVD method."). Aside from this, the specification is likewise devoid of any guidance in defining "refractory metal."

Turning to the prosecution history, Samsung argues that the Examiner cited the Takeuchi reference which discloses titanium as a refractory metal. (Feb. 16, 1996 Amendment at 3-4). The Takeuchi reference provides:

In this embodiment, the refractory metal film disposed on the polysilicon film is composed of molybdenum. The same effects can, however, be expected even by employing tungsten, titanium, platinum, cobalt, nickel, or tantalum; and refractory metal silicide films of those metals may also be used.

U.S. Patent No. 5,097,300, col. 6, ll. 35-40. Samsung further refers to the European patent EP 0 497 596 B1 issued to Liou which was also cited during prosecution. This reference likewise identifies titanium as a refractory metal. It provides:

After formation of the source/drain regions 28, a layer 30 of refractory metal preferably titanium, is formed over the surface of the device.

Liou Patent, col. 5, ll. 5-8.

Based on the intrinsic evidence, particularly the prosecution history, the Court concludes that a person of ordinary skill in the art when reading the '048 Patent would consider titanium to be a refractory metal. Thus, the Court rejects MEI's proposition that the limitation of "in excess of

2000°C” should be included in the instruction. The Court does not find, however, a proper basis for adopting Samsung’s construction that would require a skilled artisan to examine a number of properties including melting point, electrical resistivity, corrosion resistance, and temperature stability, when the intrinsic evidence hardly mentions, if at all, any of these chemical properties. As such, the Court adopts a modified construction which is based on the conclusion that titanium is included as a refractory metal and the parties’ agreement that a refractory metal has a high melting point. (Samsung Patent Chart at 2). Consequently, the term “refractory metal” means “a metal with a high melting point including titanium.”

CONCLUSION

For the reasons set forth above, the disputed claim terms have the following meanings:

A. The '998 Patent

1. the corresponding structure for "means for reading" in Claims 1, 4 and 7 that performs the claimed function "to read data from a plurality of the memory cells connected to a specific word line into a plurality of data lines by activating the specific word line of the memory cell array" is the word line driver 3 and memory cells 13;
2. the sense amplifier consists of a plurality of differential amplifiers which may sense from more than one memory cell array;
3. "sense amplifier driver" means "a circuit that is connected to a restore signal line and a drive signal line which provides voltages for operating the differential amplifiers";
4. "restore signal line" means "a line that is connected to both the sense amplifier driver and the differential amplifier through which an electric current, namely a restore signal, flows which causes the differential amplifiers to operate";
5. "drive signal line" means "a line that is connected to both the sense amplifier driver and the differential amplifier through which an electric current, namely a drive signal, flows which causes the differential amplifiers to operate";
6. "opposite ends" means "toward the termination point of the conductor lines";

B. The '095 Patent

7. "a plurality of unit memory cell blocks distributed through said memory array region at regular spacings" means "a regular distribution of a plurality of unit memory cell

- blocks in at least one direction”;
8. “a plurality of unit sense amplifier blocks distributed among said unit memory cell blocks at regular spacings” means “a regular distribution of a plurality of unit sense amplifier blocks in at least one direction”;
 9. “a plurality of sense amplifier drive circuits, distributed among said unit sense amplifier blocks at regular spacings” means “a regular distribution of a plurality of sense amplifier circuits in at least one direction”;
 10. “an upper layer of a substrate” means “a layer, situated above a lower layer, which is part of the substrate”;
 11. “a lower layer of said substrate” means “a layer, situated below an upper layer, which is part of the substrate”;

C. The ‘648 Patent

12. the corresponding structures for the “memory replacement means” in Claims 1 and 4 as disclosed in Embodiments I and III are the row predecoder 5, along with the redundancy-use decision circuit 20, the redundancy encoder 8, and the decode signal selection circuit 9 in Figures 1 and 7;
13. Embodiment II is not inoperative; the corresponding structures for the “memory replacement means” in Claims 1 and 4 as disclosed in Embodiment II is the redundancy-use decision circuit 20, the redundancy encoder 8, the circuit for generating the primary memory stop signal 18, and the memory cell block selection switching circuit 29 in Figure 6;

D. The '048 Patent

14. "edge portion" is "the two-dimensional border or boundary of the layer";
15. "recessed edge portion" requires the entire edge of the layer to be indented in its entirety;
16. "formed on" requires direct contact between the layer and the element following the term in the claim;
17. "layer" means "a thickness of material laid on, spread over, or otherwise covering a surface; a stratum";
18. "gate-insulating layer" is one particular layer that may consist of multiple layers;
19. the "gate structure having . . ." term means that the gate structure consists of a first conductive layer of polycrystalline silicon and a second conductive layer of refractory metal silicide only, and may not include additional layers;
20. "refractory metal" means "a metal with a high melting point including titanium."

Dated: March 10, 2006

s/ Garrett E. Brown, Jr.
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